CHAPTER 4

Nuclear Forces

4.1 Introduction

Up to now we have only taken into account the forces acting between nucleons in a very qualitative way. We have used some of their properties —such as their short range and saturation character—to explain the volume and surface terms in the liquid drop model. We have also assumed that they give rise to an average single-particle potential. By adjusting a few parameters, we are able to reproduce a large quantity of experimental data. The success of these phenomenological models gives us the confidence to go a step further and investigate the nuclear many-body problem from a more microscopic point of view. In particular, we wish to apply the techniques of modern many-body theory.

The starting point for all these considerations is obviously the two-body interaction between nucleons. There are three basic assumptions in this concept:

(i) Dynamical mesonic degrees of freedom can be neglected and the nucleus can be described as a system of A nucleons whose interaction can be represented by a potential.
(ii) Relativistic effects are negligible.
(iii) Only two-body forces are important.

Even with these rather drastic assumptions, we immediately run into two difficulties when we try to proceed in the way we have discussed:

(i). There exists no derivation of the *nucleon–nucleon force* from first principles. Though this should be possible in principle with the modern theory of gauge fields for quarks and gluons, attempts in this direction are only in their infancy [De 78]. On the other hand, theories that start out from an effective Lagrangian for interacting mesons and nucleons have recently proved quite successful (see, for example, [CLL 73, LLR 75, DSB 77, Vi 78, Ho 80]). The basic ingredient is the pion–nucleon coupling constant, which is known from experiment. The nucleon–nucleon force is obtained without a free parameter for particle distances greater than 0.8 fm. The part from 0 to 0.8 fm is represented by a phenomenological potential containing six parameters in each isospin channel. Excellent fits to the measured nucleon–nucleon phase shifts are achieved.

The potentials used until now have been almost entirely phenomenological (besides the Yukawa part resulting from the one-pion exchange; see below) and contain up to about 50 parameters. The experimental phase shifts are also very well reproduced with these potentials. On the basis of these forces, which we shall discuss very briefly in Section 4.2.2, we should be able to apply the method of many-body theory and to derive the phenomenological properties discussed in the preceding chapters in a quantitative way.

(ii). There is, however, a second difficulty in nuclear theory. These bare nuclear forces are, from a numerical point of view, very ill behaved. They show strong repulsion at short distances (hard core) and cannot be treated straightforwardly by the usual many-body techniques. For instance, they are too strong to be treated by perturbation theory and the hard core makes a direct self-consistent field approach (see Chap. 5), for example, impossible. In fact, the nucleons within a nucleus do not feel the bare nucleon–nucleon interaction. Taking into account that they interact with one another in the presence of many other nucleons permits one to introduce an *effective nucleon–nucleon interaction*, which is rather well behaved and allows application of the usual many-body methods, such as Hartree–Fock theory (Chap. 5). Much work has been done to derive this effective interaction from the bare nucleon–nucleon force. We will in Sec. 4.3 discuss the basic ideas that have been used to achieve this goal. As a result, a great deal has been learned about the properties and the structure of these effective forces. However, in order to reproduce experimental data quantitatively one still needs phenomenological renormalization parameters, and from this point of view the theory is not yet very satisfying.

In most of the so-called microscopic descriptions of the nucleus one uses *phenomenological effective forces*, which are constructed on the basis of these considerations, but depend on some parameters that are adjusted to fit experimental data.

In this chapter we do not want to go into such attempts to derive the bare nucleon–nucleon force [BJ 76a]. In the second section we will discuss

some invariance principles that should be obeyed by the bare forces and which already allow one to draw some conclusions about their analytical structure. In the third section, we briefly present the microscopic description of effective interactions and discuss their properties and their field of application. The fourth section presents a number of phenomenological potentials that have been used to represent the residual interaction between the nucleons moving in a given average potential.

We wish to emphasize that in this chapter we are only dealing with the nuclear interaction. The *Coulomb interaction* has to be treated separately. When a comparison with experimental data is required one has always to subtract Coulomb effects first.

4.2 The Bare Nucleon–Nucleon Force

4.2.1 General Properties of a Two-Body Force

The most general quantum mechanical two-body potential V is completely specified by its matrix elements between two-body states (in a coordinate representation $|\mathbf{r}_1, s_1, t_1; \mathbf{r}_2, s_2, t_2\rangle$; where $s_i = \pm\frac{1}{2}$ and $t_i = \pm\frac{1}{2}$ are spin and isospin coordinates) as:

$$\langle \mathbf{r}_1' s_1' t_1' \mathbf{r}_2' s_2' t_2' | V | \mathbf{r}_1 s_1 t_1 \mathbf{r}_2 s_2 t_2 \rangle. \tag{4.1}$$

The space of two-particle states $|\mathbf{r}_1, s_1, t_1; \mathbf{r}_2, s_2, t_2\rangle$ is a product space of the coordinate wave functions $|\mathbf{r}_1\rangle$ and $|\mathbf{r}_2\rangle$ and the spin and isospin vectors $|s_1\rangle$, $|s_2\rangle$ and $|t_1\rangle$, $|t_2\rangle$. Since any operator in the spin space of one particle can be represented as a linear combination of the spin matrices $\sigma_1, \sigma_2, \sigma_3$ and the unity matrix $\sigma_0 = 1$, the most general form of the operator V is

$$V = \sum_{i,k=0}^{3} V_{ik} \sigma_i^{(1)} \sigma_k^{(2)}. \tag{4.2}$$

The function V_{ik} also depends analogously on the isospin operators $\tau^{(1)}$ and $\tau^{(2)}$. In addition to this isospin dependence, the V_{ik} are, in general, integral operators in coordinate space

$$V|\mathbf{r}_1\mathbf{r}_2\rangle = \int V(\mathbf{r}_1', \mathbf{r}_2', \mathbf{r}_1, \mathbf{r}_2) |\mathbf{r}_1'\mathbf{r}_2'\rangle \, d^3r_1' \, d^3r_2'. \tag{4.3}$$

In the special case in which $V(\mathbf{r}_1', \mathbf{r}_2', \mathbf{r}_1, \mathbf{r}_2)$ has the form

$$V(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_1', \mathbf{r}_2') = \delta(\mathbf{r}_1 - \mathbf{r}_1') \delta(\mathbf{r}_2 - \mathbf{r}_2') V(\mathbf{r}_1, \mathbf{r}_2) \tag{4.4}$$

V is called a *local potential*, and we have

$$V|\mathbf{r}_1\mathbf{r}_2\rangle = V(\mathbf{r}_1, \mathbf{r}_2)|\mathbf{r}_1\mathbf{r}_2\rangle. \tag{4.5}$$

In this case the interaction between the two particles depends only on the points $\mathbf{r}_1$ and $\mathbf{r}_2$ (and eventually on the spin and isospin). It does not, for instance, depend on the velocity of the particles.

We shall show that, in general, nonlocal potentials correspond to a velocity dependence. We therefore expand*

$$|\mathbf{r}_1', \mathbf{r}_2'\rangle = |\mathbf{r}_1\mathbf{r}_2\rangle + (\mathbf{r}_1' - \mathbf{r}_1)\frac{\partial}{\partial \mathbf{r}_1}|\mathbf{r}_1\mathbf{r}_2\rangle + (\mathbf{r}_2' - \mathbf{r}_2)\frac{\partial}{\partial \mathbf{r}_2}|\mathbf{r}_1\mathbf{r}_2\rangle + \cdots$$

$$=: \exp\left\{(\mathbf{r}_1' - \mathbf{r}_1)\frac{\partial}{\partial \mathbf{r}_1} + (\mathbf{r}_2' - \mathbf{r}_2)\frac{\partial}{\partial \mathbf{r}_2}\right\} : |\mathbf{r}_1\mathbf{r}_2\rangle \tag{4.6}$$

and get, from (4.3),

$$V|\mathbf{r}_1\mathbf{r}_2\rangle = \int V(\mathbf{r}_1', \mathbf{r}_2', \mathbf{r}_1, \mathbf{r}_2)\exp\left\{\frac{i}{\hbar}(\mathbf{r}_1' - \mathbf{r}_1)\mathbf{p}_1 + \frac{i}{\hbar}(\mathbf{r}_2' - \mathbf{r}_2)\mathbf{p}_2\right\}|\mathbf{r}_1\mathbf{r}_2\rangle\, d^3r_1'\, d^3r_2'$$

$$= \tilde{V}(\mathbf{r}_1, \mathbf{p}_1, \mathbf{r}_2, \mathbf{p}_2)|\mathbf{r}_1\mathbf{r}_2\rangle. \tag{4.7}$$

This means that the most general potential can be represented by Eq. (4.2) where the V_{ik} are operators in coordinate space of the form (4.7) (for reasons of simplicity we neglected the isospin dependence).

In the following we investigate the symmetry properties of such potentials $V(1,2) = V(\mathbf{r}_1, \mathbf{p}_1, \boldsymbol{\sigma}^{(1)}, \boldsymbol{\tau}^{(1)}, \mathbf{r}_2, \mathbf{p}_2, \boldsymbol{\sigma}^{(2)}, \boldsymbol{\tau}^{(2)})$. The form of this general function can, however, be restricted by requiring the imposition of a number of symmetries.

In particular, we require the following eight symmetries:

(i) Hermeticity

(ii) Invariance under an exchange of the coordinates

$$V(1,2) = V(2,1). \tag{4.8}$$

This property is strongly connected with the symmetry of the two-particle wave function $|12\rangle$. Since nucleons are fermions, they have to be totally antisymmetric. For example, if we take a product wave function built out of ordinary space, a spin and an isospin part

$$\langle \mathbf{r}_1 s_1 t_1, \mathbf{r}_2 s_2 t_2 \,|\, 12\rangle = \varphi(\mathbf{r}_1, \mathbf{r}_2)\chi(s_1, s_2)\zeta(t_1, t_2) \tag{4.9}$$

we have four combinations compatible with the Pauli principle, which are characterized by the symmetry of the coordinate space and spin part (Table 4.1). The isospin component is determined in each case by requiring the antisymmetry of the total wave function (4.9).

*: : means *normal ordering*, i.e., the derivatives $\partial/\partial \mathbf{r}_i$ should not act on the coordinates $\mathbf{r}_i$ in the expansion of the exponent.

Table 4.1 Characterization of the symmetries of the two-particle state (4.9)

φ	χ	abbreviation	ζ
even	singlet	es	+
even	triplet	et	−
odd	singlet	os	−
odd	triplet	ot	+

(iii) Translational invariance. The potential depends on the relative coordinate $\mathbf{r}=\mathbf{r}_1-\mathbf{r}_2$ only

$$V(1,2)=V(\mathbf{r},\mathbf{p}_1,\boldsymbol{\sigma}^{(1)},\boldsymbol{\tau}^{(1)},\mathbf{p}_2,\boldsymbol{\sigma}^{(2)},\boldsymbol{\tau}^{(2)}). \tag{4.10}$$

(iv) Galilean invariance. The potential is not changed by a transformation to a system which moves with constant velocity, that is, it depends only on the relative momentum $\mathbf{p}=\frac{1}{2}(\mathbf{p}_1-\mathbf{p}_2)$:

$$V(1,2)=V(\mathbf{r},\mathbf{p},\boldsymbol{\sigma}^{(1)},\boldsymbol{\tau}^{(1)},\boldsymbol{\sigma}^{(2)},\boldsymbol{\tau}^{(2)}). \tag{4.11}$$

(v) Invariance under space reflection. Contrary to the weak interaction, there is no parity violation for strong interactions:

$$V(\mathbf{r},\mathbf{p},\boldsymbol{\sigma}^{(1)},\boldsymbol{\tau}^{(1)},\boldsymbol{\sigma}^{(2)},\boldsymbol{\tau}^{(2)})=V(-\mathbf{r},-\mathbf{p},\boldsymbol{\sigma}^{(1)},\boldsymbol{\tau}^{(1)},\boldsymbol{\sigma}^{(2)},\boldsymbol{\tau}^{(2)}). \tag{4.12}$$

(vi) Time reversal invariance guarantees that the equations of motion do not depend on the direction in which the time evolves (for details, see [Me 61])

$$V(\mathbf{r},\mathbf{p},\boldsymbol{\sigma}^{(1)},\boldsymbol{\tau}^{(1)},\boldsymbol{\sigma}^{(2)},\boldsymbol{\tau}^{(2)})=V(\mathbf{r},-\mathbf{p},-\boldsymbol{\sigma}^{(1)},\boldsymbol{\tau}^{(1)},-\boldsymbol{\sigma}^{(2)},\boldsymbol{\tau}^{(2)}). \tag{4.13}$$

(vii) Rotational invariance in coordinate space. Rotations in three-dimensional coordinate space act not only on the vectors $\mathbf{r}$ and $\mathbf{p}$ but also on the spin matrices $\boldsymbol{\sigma}=2\cdot\mathbf{s}$. With respect to spin, the operator V has the form (4.2). It has to be a scalar under a rotation in coordinate space, which means in particular that V_{00} has to be a scalar. There exist three independent scalars which we can construct from the two vectors $\mathbf{r}$ and $\mathbf{p}$, namely $\mathbf{r}^2,\mathbf{p}^2$ and $\mathbf{r}\cdot\mathbf{p}+\mathbf{p}\cdot\mathbf{r}$. However, the latter expression can only appear quadratically because of time reversal invariance (vi). It is more convenient to express $(\mathbf{rp}+\mathbf{pr})^2$ through $\mathbf{r}^2,\mathbf{p}^2$ and $\mathbf{L}^2=(\mathbf{r}\times\mathbf{p})^2$. V_{00} can then be written as a function of $\mathbf{r}^2,\mathbf{p}^2$ and $\mathbf{L}^2$. Because of invariance (ii) and (v) we find

$$V(\mathbf{r},\mathbf{p},\boldsymbol{\sigma}^{(1)},\boldsymbol{\sigma}^{(2)})=V(\mathbf{r},\mathbf{p},\boldsymbol{\sigma}^{(2)},\boldsymbol{\sigma}^{(1)}). \tag{4.14}$$

The terms in (4.2) that are linear in $\boldsymbol{\sigma}^{(i)}$ therefore depend only on

$$\mathbf{S}=\tfrac{1}{2}(\boldsymbol{\sigma}^{(1)}+\boldsymbol{\sigma}^{(2)}). \tag{4.15}$$

To form a scalar, $\mathbf{S}$ has to be multiplied by a vector, which is invariant under space reflection. Only $\mathbf{L}$ fulfils this requirement

$$\mathbf{L}\cdot\mathbf{S}=\tfrac{1}{2}\mathbf{L}(\boldsymbol{\sigma}^{(1)}+\boldsymbol{\sigma}^{(2)}) \tag{4.16}$$

The quadratic terms in $\boldsymbol{\sigma}$ in Eq. (4.2) form a tensor. It can be decomposed into a scalar $\boldsymbol{\sigma}^{(1)}\cdot\boldsymbol{\sigma}^{(2)}$, a vector $\boldsymbol{\sigma}^{(1)}\times\boldsymbol{\sigma}^{(2)}$, and a symmetric tensor with vanishing trace $(\sigma_i^{(1)}\sigma_k^{(2)}+\sigma_k^{(1)}\sigma_i^{(2)})(1-\frac{1}{3}\delta_{ik})$. Because of (4.14), $\boldsymbol{\sigma}^{(1)}\times\boldsymbol{\sigma}^{(2)}$ cannot appear. As shown by Okubo and Marshak [OM 58], the only possible independent combinations are

$$\begin{aligned}&\boldsymbol{\sigma}^{(1)}\boldsymbol{\sigma}^{(2)},\ (\mathbf{r}\boldsymbol{\sigma}^{(1)})(\mathbf{r}\boldsymbol{\sigma}^{(2)}),\ (\mathbf{p}\boldsymbol{\sigma}^{(1)})(\mathbf{p}\boldsymbol{\sigma}^{(2)}),\\&(\mathbf{L}\boldsymbol{\sigma}^{(1)})(\mathbf{L}\boldsymbol{\sigma}^{(2)})+(\mathbf{L}\boldsymbol{\sigma}^{(2)})(\mathbf{L}\boldsymbol{\sigma}^{(1)}).\end{aligned} \tag{4.17}$$

Each of these terms can be multiplied by an arbitrary function of $\mathbf{r}^2,\mathbf{p}^2$ and $\mathbf{L}^2$.

(viii) Rotational invariance in isospin space. Within the isospin formalism, protons and neutrons are considered as quantum states of one elementary particle that

form a doublet (see Sec. 2.6.3) with isospin $\frac{1}{2}$. The two-dimensional representation of the rotational group reproduces all their transformation properties. Rotations within the isospin space (as long as they are not around the 3-axis) produce mixtures of protons and neutrons. Rotational invariance of the nuclear force therefore means the same as charge independence, that is, the proton–proton interaction has the same strength as the neutron–neutron interaction. This has been confirmed by nucleon–nucleon scattering experiments as well as by the symmetry properties of mirror nuclei (e.g., He^3 and H^3). Mathematically speaking, this means that the nucleon–nucleon interaction $V(1,2)$ commutes with the operators of the total isospin

$$\mathbf{T}=\mathbf{t}_1+\mathbf{t}_2. \tag{4.18}$$

Eigenstates can then be constructed of $\mathbf{T}^2, \mathbf{T}_3$ with eigenvalues $T=0$, $T_3=0$ and $T=1$, $T_3=-1, 0, +1$ (isospin singlet and isospin triplet). Charge invariance means that $\mathbf{T}^2$ commutes with the operator of the nuclear force. Therefore, the interactions in $T=1$ states have to be the same (*pp*, *nn*, or symmetric *pn* states). However, they may be different in the $T=0$ state (antisymmetric *pn* system).

The formalism of isospin matrices $\tau=(\tau_1, \tau_2, \tau_3)$ is identical to that of regular spin. Since there is no other vector in isospin space, the only isospin invariant combination of the isospin matrices corresponding to particle 1 and 2 is

$$V_0+V_\tau \tau^{(1)}\tau^{(2)}. \tag{4.19}$$

The functions V_0 and V_τ depend on the remaining coordinates such as $\mathbf{r}, \mathbf{p}, \sigma_1, \sigma_2$, as we have already discussed.

Not all of the combinations possible from the symmetric point of view have been used to describe the nuclear force. We will mention here only the most important terms:

(i). Among the local forces, which do not depend on the velocity, the *central force* is the most important. It depends only on the distance r between the nucleons:

$$V_C(1,2)=V_0(r)+V_\sigma(r)\sigma^{(1)}\sigma^{(2)}+V_\tau(r)\tau^{(1)}\tau^{(2)}+V_{\sigma\tau}(r)\sigma^{(1)}\sigma^{(2)}\tau^{(1)}\tau^{(2)}. \tag{4.20}$$

(ii). The only remaining local part is the *Tensor force*

$$V_T(1,2)=\left[V_{T_0}(r)+V_{T_\tau}(r)\tau^{(1)}\tau^{(2)}\right]\cdot S_{12} \tag{4.21}$$

with

$$S_{12}=\frac{3}{r^2}(\sigma^{(1)}\mathbf{r})(\sigma^{(2)}\mathbf{r})-\sigma^{(1)}\sigma^{(2)}.$$

The term $-\sigma^{(1)}\sigma^{(2)}$ is added to make sure that an average of $V_T(1,2)$ taken over all directions of $\mathbf{r}$ vanishes:

$$\int V_T(1,2)\,d\Omega=0 \quad \text{where} \quad \mathbf{r}=(r,\Omega). \tag{4.22}$$

An experimental hint of the existence of a tensor component in the

nucleon–nucleon potential is given by the quadrupole moment of the deuteron, which cannot be explained by pure central forces.

(iii). The most important nonlocal term is the *two-body spin orbit interaction*

$$V_{LS} = V_{LS}(r)\mathbf{L}\cdot\mathbf{S}. \tag{4.23}$$

As we shall see in Chap. 5, such a two-body spin orbit potential causes the one-body spin orbit term in the average single-particle nuclear potential, used to explain the magic numbers in nuclei.

(iv). One sometimes also uses a *second-order spin orbit interaction*:

$$V_{LL} = V_{LL}(r)\Big\{(\boldsymbol{\sigma}^{(1)}\boldsymbol{\sigma}^{(2)})\mathbf{L}^2 - \frac{1}{2}\big[(\boldsymbol{\sigma}^{(1)}\mathbf{L})(\boldsymbol{\sigma}^{(2)}\mathbf{L}) + (\boldsymbol{\sigma}^{(2)}\mathbf{L})(\boldsymbol{\sigma}^{(1)}\mathbf{L})\big]\Big\}. \tag{4.24}$$

4.2.2 The Structure of the Nucleon–Nucleon Interaction

The central force (4.20) is the most important part of the nucleon–nucleon interaction. It can also be represented in terms of exchange or projection operators.

The operators

$$P^{\sigma} = \frac{1}{2}(1 + \boldsymbol{\sigma}^{(1)}\boldsymbol{\sigma}^{(2)}), \qquad P^{\tau} = \frac{1}{2}(1 + \boldsymbol{\tau}^{(1)}\boldsymbol{\tau}^{(2)}) \tag{4.25}$$

exchange the spin and isospin coordinates, respectively, in a wave function. For instance, we can apply P^{σ} to the wave function (4.9) and obtain

$$P^{\sigma}\varphi(\mathbf{r}_1\mathbf{r}_2)\chi(s_1, s_2)\zeta(t_1, t_2) = \varphi(\mathbf{r}_1, \mathbf{r}_2)\chi(s_2, s_1)\zeta(t_1, t_2). \tag{4.26}$$

This is easy to understand by using the operator of the total spin $\mathbf{S}$ (4.15). The eigenstates of $\mathbf{S}^2$ are singlet and triplet states and we find:

$$P^{\sigma} = \frac{1}{2}\big(1 + 2(\mathbf{S}^2 - \mathbf{s}^{(1)^2} - \mathbf{s}^{(2)^2})\big) = S(S+1) - 1 = \begin{cases} 1 & \text{for triplet} \\ -1 & \text{for singlet.} \end{cases} \tag{4.27}$$

We can also define an operator P^{r} which exchanges the spacial coordinates $\mathbf{r}_1$ and $\mathbf{r}_2$ of the particles.* Since the wave function has to be antisymmetric under the exchange of all coordinates of the particles 1 and 2, the Pauli principle may be written in the form

$$P^{r}P^{\sigma}P^{\tau} = -1. \tag{4.28}$$

We can therefore express the operator $P^{\tau} = -P^{r}P^{\sigma}$ and eliminate the products $\boldsymbol{\sigma}^{(1)}\boldsymbol{\sigma}^{(2)}$ and $\boldsymbol{\tau}^{(1)}\boldsymbol{\tau}^{(2)}$ in Eq. (4.20).

*The operator P^{r} can be represented by a nonlocal operator in coordinate space, viz:

$$V(\mathbf{r})P^{r}\psi(\mathbf{r}_1, \mathbf{r}_2) = \int V(\mathbf{r}_1 - \mathbf{r}_2)\delta(\mathbf{r}'_1 - \mathbf{r}_2)\delta(\mathbf{r}'_2 - \mathbf{r}_1)\psi(\mathbf{r}'_1, \mathbf{r}'_2)\, d^3r'_1\, d^3r'_2.$$

In this sense only Wigner forces (4.30) are local.

Finally, we obtain

$$V_C = V_W(r) + V_M(r)P^r + V_B(r)P^\sigma + V_H(r)P^rP^\sigma \tag{4.29}$$

with the following relations

$$\begin{aligned} V_W &= V_0 - V_\sigma - V_\tau + V_{\sigma\tau} && \text{(Wigner force)} \\ V_M &= -4V_{\sigma\tau} && \text{(Majorana force)} \\ V_B &= 2V_\sigma - 2V_{\sigma\tau} && \text{(Bartlett force)} \\ V_H &= -2V_\tau + 2V_{\sigma\tau} && \text{(Heisenberg force).} \end{aligned} \tag{4.30}$$

The names of these different components of the nuclear force go back to the years following 1930, when the first models of the nucleus were introduced and the saturation property of nuclear forces was explained by exchange terms without introducing a hard core (for a historical review of this work see [Br 65a]).

A third way of representing the central force uses projection operators

$$\begin{aligned} \Pi_s^\sigma &= \frac{1}{2}(1 - P^\sigma), \quad \Pi_t^\sigma = \frac{1}{2}(1 + P^\sigma), \\ \Pi_s^\tau &= \frac{1}{2}(1 - P^\tau), \quad \Pi_t^\tau = \frac{1}{2}(1 + P^\tau), \\ \Pi_o^r &= \frac{1}{2}(1 - P^r), \quad \Pi_e^r = \frac{1}{2}(1 + P^r). \end{aligned} \tag{4.31}$$

These are projection operators ($P^2 = P, P^+ = P$), which project onto the singlet (s) and triplet (t), and onto the even (e) and odd (o) part of the nuclear two-body wave function (4.9) in the sense of Table 4.1. We can express the exchange operators P^σ, P^r by these projection operators and obtain

$$V(1,2) = V_{et}(r)\Pi_e^r\Pi_t^\sigma + V_{es}(r)\Pi_e^r\Pi_s^\sigma + V_{ot}(r)\Pi_o^r\Pi_t^\sigma + V_{os}(r)\Pi_o^r\Pi_s^\sigma. \tag{4.32}$$

This representation is especially useful in practical applications; for instance, in p–p scattering experiments we have only isospin triplet states, (i.e., only es and ot are important). Table 4.2 shows those functions obtained if one operates with the different representations on wave functions with different symmetry.

Table 4.2 Connections between the different representations of a central force

$\sigma^{(1)}\sigma^{(2)}$	$\tau^{(1)}\tau^{(2)}$	$V(\sigma^{(1)}\sigma^{(2)}, \tau^{(1)}\tau^{(2)})$	$V(P^r, P^\sigma)$	$V(\Pi^r, \Pi^\sigma)$	$\|12\rangle$
1	−3	$V_{00} + V_{0\sigma} - 3V_{0\tau} - 3V_{0\sigma\tau}$	$V_W + V_M + V_B + V_H$	V_{et}	$\|et\rangle$
−3	1	$V_{00} - 3V_{0\sigma} + V_{0\tau} - 3V_{0\sigma\tau}$	$V_W + V_M - V_B - V_H$	V_{es}	$\|es\rangle$
1	1	$V_{00} + V_{0\sigma} + V_{0\tau} + V_{0\sigma\tau}$	$V_W - V_M + V_B - V_H$	V_{ot}	$\|ot\rangle$
−3	−3	$V_{00} - 3V_{0\sigma} - 3V_{0\tau} + 9V_{0\sigma\tau}$	$V_W - V_M - V_B + V_H$	V_{os}	$\|os\rangle$

The *radial dependence* of the functions V cannot be deduced from invariance principles. In 1937, Yukawa proposed an explanation of the nuclear force using a meson field theory. The nucleons influence each other by the exchange of one or several mesons. The simplest form is the one-pion exchange potential (OPEP). It has the radial dependence of the

Yukawa potential [Yu 35]

$$V_Y(r)=\frac{e^{-\mu r}}{\mu r}, \tag{4.33}$$

where $1/\mu=\hbar/m_\pi c$ is the Compton wavelength of the pion. The asymptotic form of this potential is uniquely determined by the properties of the pion and its coupling strength to the nucleonic field $g^2/\hbar c\simeq 0.081$:

$$V^{\mathrm{OPEP}}=\frac{g^2}{3\hbar c}m_\pi c^2\frac{e^{-\mu r}}{\mu r}(\tau^{(1)}\tau^{(2)})\left\{\sigma^{(1)}\sigma^{(2)}+\left(1+3\frac{1}{\mu r}+3\left(\frac{1}{\mu r}\right)^2\right)S_{12}\right\}. \tag{4.34}$$

A phase shift analysis of the nucleon–nucleon scattering data shows that the OPEP-potential (4.34) is well able to reproduce the phase shifts for large orbital angular momenta $L\geqslant 6$ [Br 67b]. Since these high partial waves only feel the tail of the nuclear force at large distances ($r\geqslant 2$ fm), we can assume that the OPEP potential describes the nuclear force properly at such large distances. For smaller distances we must, in addition, also introduce the two-pion exchange and the ρ- and ω-meson exchange in order to obtain the medium-range part of the force. This has been achieved very successfully [BJ 76a, CLL 73, LLR 75, DSB 77, Vi 78, Ho 80], so that only the short-range part of the force still has to be fitted by a phenomenological ansatz. Only six parameters are needed for each isospin state. As we mentioned in the introduction, this potential is not used very much as yet, therefore phenomenological counterparts have been employed until now. These phenomenological parametrizations consist of combinations of central, tensor, spin orbit, and higher terms, and more or less arbitrary radial functions containing up to 50 parameters, which are fitted to the experimental scattering phase shifts and to the deuteron data. There are attractive and repulsive components. At large distances they go over into the OPEP-potential, whereas at short distances they have an extremely repulsive core. Several authors have therefore used a *hard core* [$V(r)=\infty$ for $r<r_c\simeq 0.4$ fm]. Others use a very repulsive core which goes to infinity only for $r\to 0$. Such potentials are called *soft core* potentials.

Examples of such realistic nucleon–nucleon potentials using a hard core are the Hamada Johnston potential [HJ 62] and the Yale potential [LHR 62]. The Tabakin potential [Ta 64] is a nonlocal potential, separable in momentum space.

The Reid soft core potential [Re 68] is also widely used. It has the structure

$$V=V_C(\mu r)+V_T(\mu r)S_{12}+V_{LS}(\mu r)\mathbf{LS}. \tag{4.35}$$

$V_C(x)$ and $V_{LS}(x)$ have the simple form

$$V_C(x)=\sum_{n=1}^{\infty}a_n\frac{e^{-nx}}{x},\qquad V_{LS}=\sum_{n=1}^{\infty}c_n\frac{e^{-nx}}{x}, \tag{4.36}$$

and $V_T(x)$ is given by

$$V_T(x)=\frac{b_1}{x}\left\{\left(\frac{1}{3}+\frac{1}{x}+\frac{1}{x^2}\right)e^{-x}-\left(\frac{k}{x}+\frac{1}{x^2}\right)e^{-kx}\right\}+\sum_{n=2}^{\infty}b_n\frac{e^{-nx}}{x}. \tag{4.37}$$

The constants are different for all values of T, S and $I \leqslant 2$. Only a_1, b_1, and c_1 are fixed in such a way that we obtain the OPEP-potential for large distances. For $I > 2$, Reid uses the OPEP-potential.

4.3 Microscopic Effective Interactions

The bare nucleon–nucleon force has—as we have already seen in the preceding section—certain features that are rather difficult to handle in practice.

There is, for instance, the hard core (or at least the very repulsive core), which would make some of the usual concepts of nuclear many-body physics extremely complicated if not inapplicable (as in the Hartree–Fock case; see Chap. 5).* This comes from the infiniteness of the matrix elements of a force with a hard core. In these theories, a way out of this situation is to use, in place of the bare interaction, a so-called effective interaction, which is itself an infinite sum of scattering processes of two nucleons in the nuclear medium. The bare interaction is then simply the Born term of this series. The object of this procedure is twofold: First, in re-summing the series one gets rid of the hard core problem, since the new interaction is well-behaved at short distances. Second, we can show that in replacing the bare interaction by its effective counterpart we have at the same time consistently summed up more of the many-body effects than if one had taken just the bare interaction.

The main fields of application are: (i) the ground state properties of nuclei, where the scattering of two nucleons within the nuclear medium has to be considered; (ii) the forces between the so-called valence nucleons; and (iii) effective forces between "particles" and "holes." There are also effective three-body forces which we will ignore in this section.

4.3.1 Brückner's G-Matrix and Bethe Goldstone Equation†

One of the most important effective interactions in nuclear physics is the so-called Brückner G-matrix [Br 55, Da 67, and references therein]. It is, for two nucleons in the nuclear medium—in a sense yet to be specified—the analogue of the scattering matrix for two nucleons in free space.

We therefore start our considerations with the Lippmann–Schwinger equation for the scattering matrix (T-matrix; see Fig. 4.1) of two particles

*In this section, we must quite often anticipate theories and methods which are only treated later in this book. This is contrary to our usual strategy, which is to avoid this situation as much as possible. As the reader will notice, however, the microscopic theory of effective interactions is not in a very satisfactory state, so we prefer to give a short survey here together with the description of phenomenological forces, rather than devote an extra chapter to it later. (See also Appendix F.)

†The discussion in this section is partially based on Gomes, Walecka, and Weisskopf [GWW 58] and the textbook of Fetter and Walecka [FW 71].

(Messiah [Me 61] Chap. XIX, Sec. 14):

$$T^E_{\mathbf{k}_1\mathbf{k}_2,\,\mathbf{k}'_1\mathbf{k}'_2} = \bar{v}_{\mathbf{k}_1\mathbf{k}_2,\,\mathbf{k}'_1\mathbf{k}'_2} + \frac{1}{2}\sum_{\mathbf{p}_1\mathbf{p}_2} \bar{v}_{\mathbf{k}_1\mathbf{k}_2,\,\mathbf{p}_1\mathbf{p}_2} \frac{1}{E-(\mathbf{p}_1^2/2m)-(\mathbf{p}_2^2/2m)+i\eta} T^E_{\mathbf{p}_1\mathbf{p}_2,\,\mathbf{k}'_1\mathbf{k}'_2}, \tag{4.38}$$

where $\mathbf{k}_1, \mathbf{k}_2$ and $\mathbf{k}'_1, \mathbf{k}'_2$ are the momenta of the incoming and outgoing particles, respectively, and E is the total scattering energy.

If we consider the scattering of two nucleons within a nuclear medium we can show (this is derived in Sec. F.4) that it makes sense to define a scattering matrix G^E analogous to that for free particles. The changes to be made for nucleons in a nucleus are almost obvious: the plane wave indices have to be changed to shell model indices, the kinetic single-particle energies figuring in the denominator of the r.h.s. of Eq. (4.38) have to be replaced by the corresponding shell model energies, and the sum over the intermediate states has to be restricted so that it does not include states below the Fermi surface. This latter feature comes from the fact that two nucleons below the Fermi surface can only scatter into states above the Fermi surface, because all other levels are occupied and are thus excluded by the Pauli principle. Therefore, we get the following equation for the G-matrix, which is usually known under the name Bethe–Goldstone equation [BG 57] (for its mathematical derivation, see Sec. F.4).

$$G^E_{ab,\,cd} = \bar{v}_{ab,\,cd} + \frac{1}{2}\sum_{\substack{m,\,n \\ >\epsilon_F}} \bar{v}_{ab,\,mn} \frac{1}{E-\epsilon_m-\epsilon_n+i\eta} G^E_{mn,\,cd}, \tag{4.39}$$

where $ab, \ldots, mn$ are shell-model indices and ϵ_F is the Fermi energy. This equation is usually represented graphically in an obvious way, as shown in Fig. 4.2. Two lines connecting two interactions represent the "propagator" $1/(E-\epsilon_n-\epsilon_m)$. (More will be explained about graphs in Chap. 8 and Appendix F.) For $E<\epsilon_F$, we can ignore the $i\eta$ in the denominator of (4.39), and in this case the G-matrix is obviously Hermitian as can be checked immediately by iterating Eq. (4.39). Equation (4.39) is also often

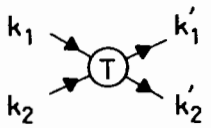


Figure 4.1. Graphical representation of the T-matrix.

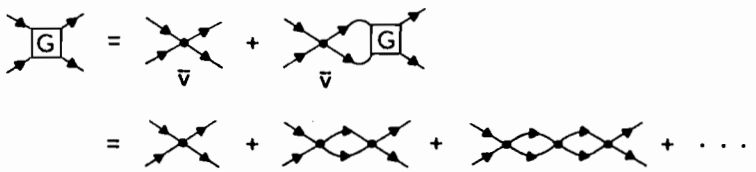


Figure 4.2. Graphical representation of the Bethe–Goldstone equation.

written in the operator form:

$$G = \bar{v} + \bar{v}\frac{Q_F}{E - H_0} G, \tag{4.40}$$

where H_0 is the shell model Hamiltonian and

$$Q_F = \sum_{\substack{m<n \\ >\epsilon_F}} |mn\rangle\langle mn| \tag{4.41}$$

is a projection operator excluding all occupied states.

In a very qualitative way, we can see from Eq. (4.40) how it can happen that G stays finite for points where $\bar{v}$ is infinite. Solving Eq. (4.40) formally yields

$$G = \frac{\bar{v}}{1 - \bar{v}Q_F/(E - H_0)}. \tag{4.42}$$

If $\bar{v}$ tends to infinity, G may stay finite. This is, of course, only a very crude argument and we shall in a definite example show in detail how this occurs. Before this, we have to discuss, however, some general features of Eq. (4.39). Despite the fact that Eq. (4.39) is formally very similar to Eq. (4.38), there are certain essential differences concerning, for instance, the boundary condition in the case of $E < \epsilon_F$ of the wave function defined in analogy to the scattered wave of two free particles by [Me 61, Chap. XIX 10]:

$$\begin{aligned} |\psi_{ab}\rangle \equiv \bar{v}^{-1} G |ab\rangle &= \left(1 + \frac{Q_F}{E - H_0} G\right) |ab\rangle \\ &= |ab\rangle + \frac{Q_F}{E - H_0} \bar{v} |\psi_{ab}\rangle, \end{aligned} \tag{4.43}$$

with

$$\langle \mathbf{r}_1 \mathbf{r}_2 \,|\, ab\rangle = \frac{1}{\sqrt{2}} (\varphi_a(\mathbf{r}_1)\varphi_b(\mathbf{r}_2) - \varphi_a(\mathbf{r}_2)\varphi_b(\mathbf{r}_1)) \tag{4.44}$$

and

$$H_0 |ab\rangle = (\epsilon_a + \epsilon_b) |ab\rangle. \tag{4.45}$$

The wave equation (4.43) formally resembles an equation for a scattered wave. However, contrary to the real scattering case, where the second term on the r.h.s. of (4.43) gives the outgoing scattered part of the wave function and thus does not vanish as the relative distance of the two particles $|\mathbf{r}_1 - \mathbf{r}_2|$ goes to infinity, in the present case of $E < \epsilon_F$, this term vanishes as $|\mathbf{r}_1 - \mathbf{r}_2|$ goes to infinity. This comes from the fact that for a real scattering process the T matrix of Eq. (4.38) enters Eq. (4.43) "on the energy shell," that is, the absolute value of the relative momenta of the two particles before and after the scattering process have to be the same; also, E must correspond to this value of the relative momentum. As a consequence, the corresponding energy denominator of Eq. (4.43) for a real scattering

process can be zero and introduce a singularity. In the present case this can never happen, since we suppose $E<\epsilon_F$ and therefore the second expression on the r.h.s. of Eq. (4.39) is never "on the energy shell," which is why it vanishes for large values of $|\mathbf{r}_1-\mathbf{r}_2|$. A derivation is presented in a review article by Day [Da 67] which, however, would be too much of a digression to be repeated here. We have thus:

$$|\psi_{ab}^{E}\rangle \xrightarrow{|\mathbf{r}_1-\mathbf{r}_2|\to\infty} |ab\rangle. \tag{4.46}$$

From Eqs. (4.43) and (4.41) we also immediately get the normalization condition for $a, b<\epsilon_F$:

$$\langle ij\,|\,\psi_{ij}\rangle = 1, \qquad i, j<\epsilon_F. \tag{4.47}$$

It is also instructive to expand the correlated pair function $|\psi_{ij}\rangle$ in an uncorrelated basis. From (4.43) we have:

$$|\psi_{ij}\rangle = |ij\rangle + \frac{1}{2}\sum_{mn} C_{mn}^{ij}|mn\rangle, \tag{4.48}$$

that is, the correlated function contains, in addition to $|ij\rangle$, components above the Fermi level. It turns out that in the most important applications of the G-matrix E lies below the Fermi level. For instance, in the Brückner Hartree–Fock theory (see Chap. 5) we must calculate $G_{ab,ij}^{E=\epsilon_i+\epsilon_j}$ with $i, j<\epsilon_F$. We will therefore treat the hard core for this case in a very simplified but explicitly solvable model which, however, shows the essential features.

Let us consider a large nucleus, the interior of which will be presumed to be not very much different from the situation where we consider an infinitely large nucleus, usually termed nuclear matter. In addition, we will make a further crude assumption, namely, that the interaction consists of a hard core only:

$$V(r)=0 \text{ for } r\geqslant c \qquad \text{and} \qquad V(r)=\infty \text{ for } r<c.$$

Furthermore, we assume that we have two distinguishable particles, for example, a proton and a neutron, which will be sufficient as a demonstration of the principle. Let us now write Eq. (4.43) in the form:

$$(E-H_0)|\psi_{ab}\rangle = (E-H_0+Q_FG)|ab\rangle, \tag{4.49}$$

which in our case ($E=\epsilon_i+\epsilon_j$) specializes to

$$(\epsilon_i+\epsilon_j-H_0)|\psi_{ij}\rangle = Q_FG|ij\rangle = Q_F\bar{v}|\psi_{ij}\rangle, \tag{4.50}$$

where for the last equality use has been made of (4.43). The general solution of (4.50) is much harder than the solution of an ordinary Schrödinger equation because of the operator Q_F, which is a nonlocal integral operator. This is best seen in the r-representation

$$\langle \mathbf{r}_1\mathbf{r}_2|Q_F|\mathbf{r}_1'\mathbf{r}_2'\rangle = \sum_{\substack{m<n\\ >\epsilon_F}} \langle \mathbf{r}_1\mathbf{r}_2\,|\,mn\rangle\langle mn\,|\,\mathbf{r}_1'\mathbf{r}_2'\rangle. \tag{4.51}$$

In nuclear matter it is clear that the single-particle wave functions that appear in (4.50) are plane waves. For simplicity we will also assume that the single-particle energies are purely kinetic energies (in a surrounding medium this will not gener-

ally be the case). Because of translational invariance, for nuclear matter the center of mass motion is trivial, and we have:

$$\langle \mathbf{r}_1\mathbf{r}_2 \,|\, \psi_{ij}\rangle \to \psi_{\mathbf{p}_1\mathbf{p}_2}(\mathbf{r}_1\mathbf{r}_2) = \frac{1}{(2\pi)^{3/2}} e^{i\mathbf{P}\mathbf{R}} \psi_{\mathbf{P}\mathbf{k}}(\mathbf{r}). \tag{4.52}$$

Here the following transformations to relative and center of mass coordinates have been made:

$$\begin{aligned} \mathbf{P} &= \mathbf{p}_1 + \mathbf{p}_2, \qquad & \mathbf{k} &= \frac{1}{2}(\mathbf{p}_1 - \mathbf{p}_2), \\ \mathbf{R} &= \frac{1}{2}(\mathbf{r}_1 + \mathbf{r}_2), & \mathbf{r} &= \mathbf{r}_1 - \mathbf{r}_2. \end{aligned} \tag{4.53}$$

With all these assumptions, Eq. (4.50) acquires the form

$$\begin{aligned} &\frac{\hbar^2}{2m}\{p_1^2 + p_2^2 + \Delta_1 + \Delta_2\}\psi_{\mathbf{p}_1\mathbf{p}_2}(\mathbf{r}_1\mathbf{r}_2) \\ &\quad = \frac{1}{(2\pi)^6}\int_{p,\, p' > k_F} d^3p\, d^3p' \int d^3r_1'\, d^3r_2' e^{i(\mathbf{p}\mathbf{r}_1 + \mathbf{p}'\mathbf{r}_2 - \mathbf{p}\mathbf{r}_1' - \mathbf{p}'\mathbf{r}_2')} v(|\mathbf{r}_1' - \mathbf{r}_2'|)\psi_{\mathbf{p}_1\mathbf{p}_2}(\mathbf{r}_1'\mathbf{r}_2'). \end{aligned} \tag{4.54}$$

Transforming this equation to relative and center of mass coordinates according to (4.53) we obtain

$$\begin{aligned} &\frac{\hbar^2}{m}\{k^2 + \Delta_r\} e^{i\mathbf{P}\mathbf{R}}\psi_{\mathbf{P}\mathbf{k}}(\mathbf{r}) \\ &\quad = \frac{1}{(2\pi)^6}\int\int_{\left|\frac{\mathcal{P}}{2} \pm \mathbf{p}\right| > k_f} d^3\mathcal{P}\, d^3p \int\int d^3R'\, d^3r'\, e^{i[\mathcal{P}\mathbf{R} + \mathbf{p}(\mathbf{r} - \mathbf{r}')]} e^{-i\mathcal{P}\mathbf{R}'} v(r') e^{i\mathbf{P}\mathbf{R}'} \psi_{\mathbf{P}\mathbf{k}}(\mathbf{r}'). \end{aligned} \tag{4.55}$$

Since $\mathcal{P}^2 = p^2 + 2pp'\cos\theta + p'^2$, we see that even under the restriction $p,\, p' > k_F$, $\mathcal{P}$ can take on all values from 0 to ∞. The integration over $\mathbf{R}'$ gives $\delta(\mathbf{P} - \mathcal{P})$ and we can therefore also perform the $\mathcal{P}$ integral. We are thus left with the following equation.

$$\frac{\hbar^2}{m}\{k^2 + \Delta_r\}\psi_{\mathbf{P}\mathbf{k}}(\mathbf{r}) = \frac{1}{(2\pi)^3}\int_{\left|\frac{\mathbf{P}}{2} \pm \mathbf{p}\right| > k_F} d^3p\, e^{i\mathbf{p}\mathbf{r}'} \int d^3r' e^{-i\mathbf{p}\mathbf{r}'} v(r')\psi_{\mathbf{P}\mathbf{k}}(\mathbf{r}'). \tag{4.56}$$

This equation is not only more complicated than a usual two-particle equation because of its integrodifferential structure, but also because the wave function has a nontrivial dependence on the center of mass momentum $\mathbf{P}$. For our purpose, it will be sufficient to evaluate it at $\mathbf{P} = 0$. Furthermore, we can decompose Eq. (4.56) into partial waves [Me 61 Chap. X, Sec. 8]:

$$\psi_{\mathbf{k}}(\mathbf{r}) = \sum_{lm} \psi_{kl}(r)\, Y_{lm}(\theta, \varphi). \tag{4.57}$$

Considering the equation for the s-wave and splitting the integral in (4.56) into two parts, $\int_{k_F}^{\infty} = \int_0^{\infty} - \int_0^{k_F}$, we obtain:

$$\begin{aligned} &\frac{\hbar^2}{m}\left(k^2 + \frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r}\right)\psi_{\mathbf{k}}(r) \\ &\quad = v(r)\psi_{\mathbf{k}}(r) - (4\pi)^2 \int_0^{k_F} \frac{dp\, p^2}{(2\pi)^3} j_0(pr) \int_0^{\infty} dr' r'^2 j_0(pr') v(r')\psi_{\mathbf{k}}(r'), \end{aligned} \tag{4.58}$$

where j_0 is the lowest-order spherical Bessel function. With $\psi_k(r)=(1/r)u_k(r)$, we finally get:

$$\frac{\hbar^2}{m}\left(k^2+\frac{\partial^2}{\partial r^2}\right)u_k(r)=v(r)u_k(r)-k_F\int_0^\infty dr'\chi(r',r)v(r')u_k(r'), \tag{4.59}$$

with

$$\chi(r,r')=\frac{2rr'}{\pi k_F}\int_0^{k_F}dp\,p^2j_0(pr)j_0(pr')=\frac{1}{\pi\cdot k_F}\left\{\frac{\sin k_F(r-r')}{r-r'}-\frac{\sin k_F(r+r')}{r+r'}\right\}. \tag{4.60}$$

It is now convenient to introduce the following dimensionless quantities:

$$x=k_F\cdot r;\qquad x'=k_F\cdot r';\qquad K=\frac{k}{k_F};\qquad \frac{mv}{k_F^2\hbar^2}=w. \tag{4.61}$$

This leads to

$$\left(\frac{d^2}{dx^2}+K^2\right)u_K(x)=w(x)u_K(x)-\int_0^\infty dx'\,\chi(x,x')w(x')u_K(x'). \tag{4.62}$$

For a square well barrier of finite height, the wave function and its first derivative are continuous at the edge of the barrier. We can be easily convinced that the first derivative of the wave function becomes discontinuous at the edge as the barrier height goes to infinity, that is:

$$u_K'(c+\epsilon)=A, \tag{4.63}$$

where A is a constant, $r=c$ is the radius of the hard core, and ϵ is an infinitesimally small positive quantity. In order to produce such a discontinuity for u of Eq. (4.62), the product $w\cdot u$ must be proportional to a δ function for $r=c$. Since for $r>c$ the potential w is zero and u finite, the product $w\cdot u$ vanishes outside the hard core radius. The wave function u cannot penetrate inside the infinite hard core ($u=0$); since there $w=\infty$, the product $w\cdot u$ may be finite and we can write, with $c'=ck_F$:

$$w(x)u_K(x)=A\delta(x-c')+l(x)\theta(c'-x). \tag{4.64}$$

The function $l(x)$ can be determined from the requirement that for $x<c'$ the left hand side of Eq. (4.62) must be zero, since $u\equiv 0$ for $x<c'$. We therefore have from Eq. (4.62) with Eq. (4.64)

$$l(x)=A\chi(x,c')+\int_0^{c'}dx'\,\chi(x,x')l(x')\qquad\text{for}\quad x<c'. \tag{4.65}$$

Since the hard core is usually rather small ($c'=0.57$ at the nuclear matter density for $c=0.4$ fm), we can develop the kernel in Eq. (4.65) and obtain from Eq. (4.60)

$$\chi(x,x')\to\frac{2xx'}{3\pi}\qquad x<c';\quad x'<c'. \tag{4.66}$$

With Eqs. (4.66) and (4.65), we see that $l(x)$ is of order c'^2, whereas any integral over $l(x)$ will be of order c'^3. In (4.62) we can therefore neglect the second term in (4.64) to obtain a result which is correct to first order in c'. We obtain

$$\begin{aligned}\left(\frac{d^2}{dx^2}+K^2\right)u_K(x)&\simeq A[\delta(x-c')-\chi(x,c')]+A\chi(x,c')\theta(c'-x)\\&=A\frac{2xc'}{\pi}\int_1^\infty dp\,p^2j_0(px)j_0(pc')+A\chi(x,c')\theta(c'-x)\\&=F(x),\end{aligned} \tag{4.67}$$

where we have made use of the identity

$$\int_0^\infty dp\, p^2 j_l(pr) j_l(pr') = \frac{\pi}{2r^2}\delta(r-r'). \tag{4.68}$$

We remark that since the r.h.s. of Eq. (4.67) is only correct to first order in c', it is somewhat arbitrary whether to include the second term of the r.h.s. of Eq. (4.67), which is of second order. The general solution of Eq. (4.67),

$$u_K(x) = \frac{\sin Kx}{K}\int_{c'}^{x} dy\, F(y)\cos Ky - \frac{\cos Kx}{K}\int_{c'}^{x} dy\, F(y)\sin Ky, \tag{4.69}$$

is obviously zero for $x = c'$. Equation (4.69) is the solution to Eq. (4.67), as can be easily verified by direct insertion.

The only unknown in the solution (4.67, 4.69) is the constant A, which we are going to determine by the requirement (4.46) that u_K has to approach asymptotically the unperturbed value. To this end we will first show that the second integral on the r.h.s. of Eq. (4.69) is zero in the limit $x \to \infty$. Since we considered our solution in the limit of very small c', we can take the integral from 0 to ∞ instead of from c' to ∞, and thus have

$$\begin{aligned}\frac{1}{K}\int_0^\infty dy\, F(y)\sin Ky &= \frac{2Ac'}{\pi}\int_1^\infty dp\; p^2 j_0(pc')\int_0^\infty dy\; y^2 j_0(Ky) j_0(py) \\ &= \frac{2Ac'}{\pi}\int_1^\infty dp\; p^2 j_0(pc')\frac{\pi}{2Kp}\delta(K-p) \\ &= 0. \end{aligned} \tag{4.70}$$

The last integral vanishes because p is outside the Fermi sphere, whereas K is inside. For $r \to \infty$ we therefore find the result that the wave function approaches the unperturbed result ($\psi \to j_0$) [see Eq. (4.46)] only if

$$\int_0^\infty dy\, F(y)\cos Ky = 1, \tag{4.71}$$

and we therefore get, using Eq. (4.67):

$$A = \left\{\cos Kc' - \int_0^\infty dy \cos(Ky)\chi(y, c')\right\}^{-1}, \tag{4.72}$$

which completes the determination of the wave function $u_K(x)$ of Eqs. (4.67) and (4.69) for a hard sphere potential.

In Fig. 4.3 we show the solution $\psi_{\mathbf{k}}(r)$ for $\mathbf{k} = 0$, which reveals several interesting features. The wave function vanishes inside the hard core. With (4.69) and (4.72) it can easily be seen that it approaches rapidly (by

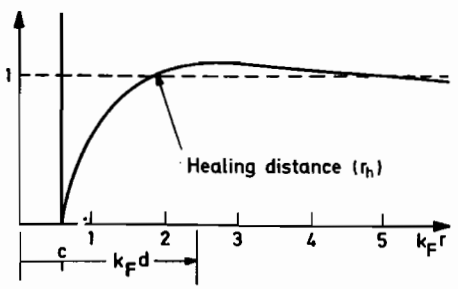


Figure 4.3. *S*-wave solution of the Bethe–Goldstone equation for a pure hard core potential in nuclear matter. For details of the figure, see text. (From [FW 71].)

damped oscillations) the value of the wave function for the noninteracting pair (which is itself equal to 1 for $k=0$ and all values of r). The r value at which the wave function first attains the unperturbed value is called the *healing distance* r_h, because it is only for distances smaller than this value that the hard core wave function differs appreciably from the unperturbed wave function. For the values chosen in our example, the healing distance is:

$$k_F r_h \simeq 1.9. \tag{4.73}$$

(This value is more or less independent of $\mathbf{P}$ and k, as a more general study of Eq. (4.59) shows [WG 67a]). It is important to note that this value is appreciably smaller than the average interparticle distance d in nuclear matter defined by the expression $1/d^3 \equiv N/V = (4\int_{p\leqslant\hbar k_F} d\mathbf{p}\int d\mathbf{r})/V(2\pi\hbar)^3 = 2k_F^3/3\pi^2$, which would yield for the interparticle distance:

$$k_F d = \left(\frac{3\pi^2}{2}\right)^{1/3} = 2.46. \tag{4.74}$$

On the average, therefore, the nucleons return after a collision to their independent particle wave function *before* the next collision takes place. A great portion of the nuclear wave function therefore consists of a determinant built out of independent particle wave functions. This can be considered as a justification of the independent particle model—the Pauli principle suppressing the low momentum components of the scattering process, that is, suppressing the long range correlations. The nucleons thus move through the nucleus most of the time as independent particles because of the exclusion principle. That this can happen in a strongly interacting but dilute Fermi gas was first pointed out by Weisskopf [We 50].

We have seen that the solution of the Bethe–Goldstone Equation (4.39) is far from trivial, and one can easily imagine that the task can become tremendously difficult for finite nuclei, where the wave functions are no longer plane waves and translational invariance is also lost. Several approximation schemes have therefore been currently applied. As we have seen in our example, it is the projection operator Q_F which makes things so difficult. It has been proposed, that the Pauli principle should be treated only perturbatively in the *separation method* of Moszkowski and Scott [MS 60] and in the *reference spectrum method* of Bethe, Brandow, and Petschek [BBP 63]. We do not wish to go into details of these approximation methods and refer the reader to the text books of Brown [Br 64], de Shalit and Feshbach [SF 74], and the review article by Bethe [Be 71], in which these and further methods, like the use of the oscillator basis for the solution of the Bethe–Goldstone equation in finite nuclei, are explained in detail. Here we want to mention only one other approximation scheme which has recently become very successful in connection with the Brückner–Hartree–Fock theory (see Chap. 5). This is the *local density approximation* by Negele [Ne 70, 75], originally introduced by Brückner, Gammel, and Weitzner [BGW 58] and Brückner, Lockett, and Rotenberg [BLR 61]. The assumption is that the G-matrix at any place in a finite nucleus is the same as that for nuclear matter at the same density, so that locally one can

calculate the G-matrix as in a nuclear matter calculation [Be 71, Ne 75]. We shall come back to this approximation in Chapter 5.

4.3.2 Effective Interactions between Valence Nucleons

Another field to which the Brückner G-matrix has important applications is the effective forces between the so-called valence nucleons (for a review, see [Ku 74b] and references therein). As a definite example, let us take ^{18}O or ^{18}F, where we have two nucleons (the "valence nucleons") on top of a doubly magic nucleus ^{16}O. In the pure shell model approximation the two nucleons will be in the $1d5/2$ level. We may hope to get a good description of the low-lying states of ^{18}O by making a configuration mixing calculation using the states $1d5/2$, $2s1/2$, $1d3/2$ which just form the first shell above the ^{16}O core. Such shell model calculations have been described to some extent in Chapter 2 and will be treated in greater detail in Chapter 8, where we will show, for instance, that such two-valence nucleons may be well described by the particle–particle Tamm–Dancoff secular equation [see Chap. 8, Eq. (8.10)]:

$$\left(E_\rho^{TD}-\epsilon_m-\epsilon_n\right)R_{mn}^\rho=\frac{1}{2}\sum_{m'n'}\bar{v}_{mn,\,m'n'}R_{m'n'}^\rho. \tag{4.75}$$

Here the ϵ_m's are the single-particle energies of the phenomenological or self-consistent (see Chap. 5) single-particle Hamiltonian H_0 and all the indices in Eq. (4.75) are in the *model space* above the Fermi level of the core (e.g., the $d5/2$, $2s1/2$, and $d3/2$ states in our example). For the two-particle interaction in Eq. (4.75), we cannot take the bare interaction, since this force has to simulate and correct at least approximately the omission of (a) the rest of the two-particle configuration above the Fermi level, and (b) contributions from higher configurations of the shell model basis like $3p$–$1h$, $4p$–$2h$, etc.

In the following we will derive an exact equation, formally almost identical to Eq. (4.75), with, however, $\bar{v}$ replaced by an effective (energy dependent) interaction, calculable at least in principle, from the bare interaction (such as, e.g., the Hamada–Johnston potential). We start out with the observation of Chapter 2 that the shell model provides a complete set of states; therefore, any exact wave function can be expanded in this basis. A convenient way to do this is given by the formalism of second quantization, in which the expansion of, for example, an $A+2$ nucleon state can be written in the following way (see also Chap. 8). (In the following, the indices m, n, r, p, and i, j shall be states above and below the Fermi level, respectively.)

$$\begin{aligned}|A+2,\tau\rangle=&\sum_{m<n}R_{mn}^\tau a_m^+a_n^+|HF\rangle+\sum_{m<n<r,\,i}R_{mnr,\,i}^\tau a_m^+a_n^+a_r^+a_i|HF\rangle\\&+\sum_{\substack{m<n<r<p\\ i<j}}R_{mnrp,\,ij}^\tau a_m^+a_n^+a_r^+a_p^+a_ia_j|HF\rangle.\end{aligned} \tag{4.76}$$

Here we assume that the ground state $|HF\rangle$ of the A nucleus in the shell model is ideally given by a Hartree–Fock calculation (see Chap. 5), but our considerations hold equally for any phenomenological shell model. The indices m, n, r, p and i, j in Eq. (4.76), of course, correspond to the shell model potential in question. The expansion consists of multiparticle–multihole components ranging from $2p$–$0h$ up to, in principle, $(A+2)p$–Ah components. We now introduce two projectors P and Q. The projector P projects on those $2p$–$0h$ states which lie within the given model space and Q projects onto the rest, that is, on those $2p$–$0h$ components which do not lie in the model space and on *all* $3p$–$1h$, $4p$–$2h$, etc. components. Therefore, $1 = P + Q$ and

$$P = \sum_{m<n}{}' a_m^+ a_n^+ |HF\rangle\langle HF| a_n a_m, \tag{4.77}$$

where the prime on the sum indicates that it goes over the model space only. We also have the usual relations for projectors, viz:

$$Q^2 = Q; \qquad P^2 = P; \qquad P^+ = P; \qquad Q^+ = Q; \qquad PQ = QP = 0. \tag{4.78}$$

With the aid of these projectors we can write for Eq. (4.76):

$$|\tau\rangle = P|\tau\rangle + Q|\tau\rangle = PR|HF\rangle + QS|HF\rangle, \tag{4.79}$$

where $R = \sum'_{m<n} R^\tau_{mn} a_m^+ a_n^+$, the prime having the same meaning as in Eq. (4.77); the operator S is then defined in an obvious way by Eqs. (4.76) and (4.79). The Schrödinger equation $(H_0 + V)|\tau\rangle = E_\tau|\tau\rangle$, where H_0 is the shell model Hamiltonian and V the residual interaction (see Chaps. 2 and 8) can now be written in the following form [Fe 62].

$$(-E_\tau + H_0 + PVP)P|\tau\rangle = -PVQ|\tau\rangle, \tag{4.80}$$

$$(-E_\tau + H_0 + QVQ)Q|\tau\rangle = -QVP|\tau\rangle. \tag{4.81}$$

Here we have multiplied the Schrödinger equation from the left once with P and once with Q and observed that P and Q commute with H_0. Solving Eq. (4.81) for $Q|\tau\rangle$ and substituting into Eq. (4.80) yields

$$\left(H_0 + PV^{E_\tau}_{\text{eff}}P\right)P|\tau\rangle = E_\tau P|\tau\rangle \tag{4.82}$$

with

$$V^{E_\tau}_{\text{eff}} = V + VQ\frac{1}{E_\tau - H_0 - QVQ}QV. \tag{4.83}$$

Multiplying Eq. (4.82) from the left with $\langle HF|a_n a_m$, where n, m are in the model space, we can write:

$$\{\langle HF|[a_n a_m, H_0] + \langle HF|a_n a_m PV_{\text{eff}}P\}\sum_{m'<n'}{}' a_{m'}^+ a_{n'}^+ |HF\rangle R^\tau_{m'n'} = (E_\tau - E_0^{HF})R^\tau_{mn} \tag{4.84}$$

with

$$H_0|HF\rangle = E_0^{HF}|HF\rangle.$$

The commutator in Eq. (4.84) is easily evaluated, and we finally obtain an

equation which is formally very similar to the two-particle Tamm–Dancoff Eq. (4.75) but is, in fact, rigorously exact:

$$\left[\left(E_\tau - E_0^{HF}\right) - \epsilon_m - \epsilon_n\right] R^\tau_{mn} = \sum_{m'<n'}{}' \langle mn | V^E_{\text{eff}} | m'n' \rangle R^\tau_{m'n'}. \tag{4.85}$$

We see that the difference between Eqs. (4.85) and (4.75) is simply that the matrix elements of the phenomenological two-body interaction have been replaced by those of an effective energy dependent interaction which (at least in principle) can be calculated from the bare interaction.* The expression (4.83) can be rewritten as an integral equation for V_{eff}:

$$V_{\text{eff}} = V + VQ \frac{1}{E - H_0} Q V_{\text{eff}}. \tag{4.86}$$

This can be verified either by expanding both (4.86) and (4.83) in powers of V, or by direct substitution of (4.83) in (4.86) and using decomposition into partial fractions. In order to investigate the simplest contributions to V_{eff}, we have to look at the structure of the projection operator Q. We can distinguish three different contributions: (i) $2p$–$0h$ excited states, where only one particle is outside the model space; (ii) two-particle excited states, where both particles of the $2p$–$0h$ components are outside the model space; (iii) those components which involve holes ($3p$–$1h$, etc.) The excited states of type (i) probably do not contribute very much, since it has been shown that their contribution vanishes exactly in nuclear matter [Ma 69], and are thus expected to be small in finite (but heavy) nuclei. Their contribution is thus omitted in practical calculations. The two-particle excited states give the most important contribution, as practical calculations have shown, therefore we want to study how they can be treated and what their relation to the Brückner G-matrix will be.

We shall call that part of Q which corresponds to the two-particle excited states (ii) Q_{2p}. Retaining in (4.86) only the Q_{2p} part and expanding in powers of V, we obtain up to second order

$$\langle mn | V_{\text{eff}} | m'n' \rangle = \bar{v}_{mn,\, m'n'} + \sum_{\substack{p<q \\ > \epsilon_{\max}}} \frac{\bar{v}_{mn,\, pq}}{\Omega_\tau - \epsilon_p - \epsilon_q} \bar{v}_{pq,\, m'n'} + \cdots, \tag{4.87}$$

where $\epsilon_{\max}$ is the upper limit of the model space and $\Omega_\tau = E_\tau^{A+2} - E_0^A$ in accordance with the footnote on this page (no problems with linked or unlinked terms appear at this level). This shows that this part of the

*The effective interaction defined by Eqs. (4.83) and (4.85) contains so-called unlinked contributions [Ma 69]. By reordering these terms in Eq. (4.85) we can show that one obtains an equation structurally equal to Eq. (4.85), where, however, E_0^{HF} is replaced by the exact ground state energy E_0^A of the core. The argument $E_\tau^{A+2} - H_0$ in the denominator of Eq. (4.83) is changed to $E_\tau^{A+2} - E_0^A - (H_0 - E_0^{HF})$ and only the linked contributions of Eq. (4.83) have to be taken into account. However, no integral equation for this quantity is known [compare Eq. (4.86)]. More details of this procedure can be found in the review articles by MacFarlane [Ma 69] and Barrett and Kirson [BK 73]. Brandow [Br 67a] has shown that we can also get rid of the energy dependence of the effective interaction by introducing the so-called "folded diagram" (see also [JB 71, EO 77]). The discussion of these rather involved techniques would go beyond the scope of this book, thus we refer the interested reader to the original literature.

effective interaction is equal to the Brückner G-matrix (4.39) with ϵ_F replaced by $\epsilon_{\max}$, that is,

$$\left(V_{\text{eff}}^{(2p)}\right)_{mn,\,m'n'} = G^{E_\tau^{A+2}-E_0^A}_{mn,m'n'}(\epsilon_F \to \epsilon_{\max}). \tag{4.88}$$

The whole intention and philosophy of rewriting the exact Schrödinger equation in the form (4.85) is based on the hope that V_{eff} is an operator which for the physical problem of two valence nucleons can be calculated to a good approximation in some perturbative way. One must therefore assume that V_{eff} is well behaved in the sense that it has, for instance, no strong energy dependence. The energy dependence of G in Eq. (4.88) is thus usually neglected and replaced by

$$\Omega_\tau \simeq \frac{1}{2}(\epsilon_m + \epsilon_n + \epsilon_{m'} + \epsilon_{n'}). \tag{4.89}$$

Also, all the other approximations commonly applied to solve the Bethe–Goldstone equation should not influence the result too much (see, for example, [Ma 69] and [SF 74]). Kuo and Brown [KB 66] have solved the Bethe–Goldstone equation in the harmonic oscillator approximation [SF 74] for ^{18}O and ^{18}F using as bare interaction the Hamada–Johnston force. This force was subsequently used in Eq. (4.85) for a diagonalization in the model space where the experimental values found in ^{17}O have been used for the single particle energies. In Fig. 4.4 we show a comparison of the

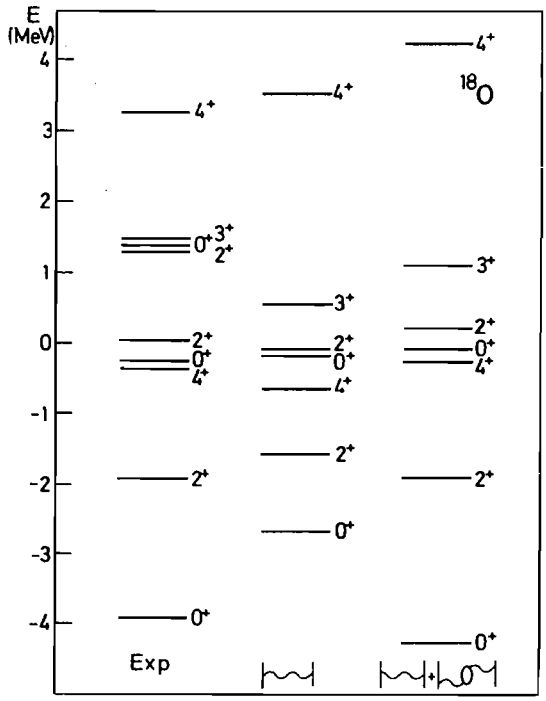


Figure 4.4. The spectrum of ^{18}O. The first column shows the experimental values. The second one was calculated with the pure G-matrix. The third also includes the polarization term (4.90). (From [HK 72].)

low-lying spectrum of ^{18}O as calculated by these authors, with the spectrum of the experiment; in Fig. 4.5 we show the same for the case of ^{210}Pb. The latter calculations were performed by Herling and Kuo [HK 72] using in principle the same method as Kuo and Brown. Their model space consisted of the following single-particle states:

neutrons: $2g9/2$ $1i11/2$ $1j15/2$ $3d5/2$ $4s1/2$ $2g7/2$ $3d3/2$
protons: $1h9/2$ $2f7/2$ $1i13/2$ $2f5/2$ $3p3/2$ $3p1/2$

As can be seen from Fig. 4.4 and Fig. 4.5, the comparison of these two calculations with experiment is not very satisfying and the agreement can be considered as qualitative only. The disagreement is probably not due to the various approximations which entered the solution of the Bethe–Goldstone equation, but rather to the omission of other configurations like $3p-1h$, etc. In both of the calculations mentioned it has been shown that at least perturbative inclusion of $3p-1h$ configurations improves agreement with experiment very much. The procedure was to calculate the contribution to V_{eff} from Q_{3p-1h} in Eqs. (4.83) and (4.85) in second order perturbation theory (omitting the unlinked terms, see footnote on page 166). This is

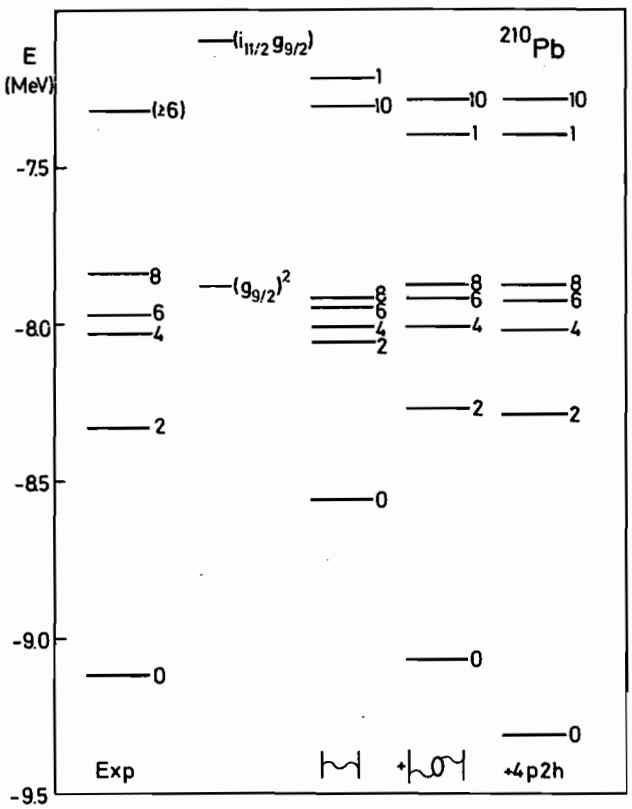


Figure 4.5. The spectrum of ^{210}Pb. The first three spectra have the same meaning as those of Fig. 4.4. In the fourth spectrum also, $4p-2h$ polarization terms were taken into account. Also indicated is the free two-particle energy. (From [HK 72].)

not too hard to carry out using our definition [(4.83) and (4.85)] and the result is:

$$\left(V_{\text{eff}}^{3p-1h}\right)_{mn,\,m'n'} = \left\{ \sum_{pi} \frac{\bar{v}_{mi,\,m'p}\bar{v}_{pn,\,in'}}{\Omega_\tau - \epsilon_{m'} - \epsilon_n - \epsilon_p + \epsilon_i} - (n' \leftrightarrow m') \right\} - \{m \leftrightarrow n\}. \tag{4.90}$$

In this expression we neglected all terms which give a renormalization of the single-particle energies, that is, terms of the structure $(V_{\text{eff}})_{mn,\,m'n'} = A_{mm'}B_{nn'} - (m' \leftrightarrow n')$, because it is believed that these terms are already included in the experimental single-particle energies used in the specific calculations. Usually one represents the term (4.90) graphically, as displayed in Fig. 4.6 (more about graphs is explained in Chap. 8 and Appendix F). The second order part however cannot be used as it stands because of the hard core potential. It can be shown [BK 67a] that it is a consistent re-summation of higher order terms to replace the bare interactions by their corresponding G-matrices in the second order contribution of the $3p$–$1h$ states to V_{eff} (e.g., (4.90)). We do not wish to go into the details of such arguments here because they would lead us outside the scope of this book. In Figs. 4.4 and 4.5 are also shown results with the inclusion of these "renormalized" second order $3p$–$1h$ contributions (the "core polarization") and, as can be seen, the agreement with experiment proves in both cases almost perfect. If this were the whole story, the results would be very satisfying, since we would have explained the low-lying part of the spectrum of these nuclei using essentially no free parameter. Unfortunately, things are far from being settled, since further studies have shown that inclusion of higher order terms will again worsen the results. This effect is shown in Fig. 4.5, where not only $3p$–$1h$ but also $4p$–$2h$ terms are included. More systematic but complicated studies in this direction have been performed by Barrett and Kirson [BK 73] and Kirson and Zamick [KZ 70], which show that great fluctuations of the results as a function of different higher order terms are observed and, therefore, no definite conclusions can be drawn; in particular, it remains unexplained why the second order $3p$–$1h$ inclusion gives such good agreement with experiment. One may, therefore, draw the conclusion that the microscopic theory of effective interactions is still not at a very satisfying stage.

Nevertheless, we want to come back to two formal points in the theory. The first concerns the energy dependence of the effective interaction V_{eff}^E. In all calculations this energy dependence has been more or less neglected. As we have already stated, this is, of course, only true as long as the energy

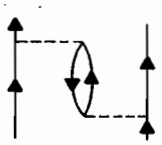


Figure 4.6. Graphical representation of a core polarization term.

dependence is really weak. However, it can happen that V_{eff}^{E} has poles, as a function of E, which lie close or even in the region of the low-lying states of interest; for example, this is the case for ^{18}O, where a collective $4p$–$2h$ pole comes very low in energy [Sch 75]. In this case, such a pole has to be treated separately, since it gives rise to a strong energy dependence in the region of interest. The states also cause difficulties in the case of the folded diagram formalism (see footnote on page 166), where they have been called *intruder states* [SW 72, 73]. There, it turns out that the perturbation series of the corresponding effective interaction in powers of the G-matrix very likely diverge in such cases. Special techniques have been developed to handle this problem [SW 72, 73, HLR 74].

Up to now we have considered only two-particle systems, however, the same considerations can be made for the case of more than two valence nucleons [Ma 69], although we do not want to go into these details.

4.3.3 Effective Interactions between Particles and Holes

The problem of effective interactions between particles and holes arises mainly in the study of vibrational states of closed shell nuclei (see Chap. 8). However, from a microscopic point of view this case has been much less investigated than the interaction between valence nucleons (see, for example, [BK 69, KBB 70]). One can argue that it would be desirable to express also the effective *ph* force by a microscopic G-matrix with perhaps some higher order corrections, like the polarization diagrams for the valence nucleons. This can, in fact, be achieved. The discussion is complicated by the fact that, for the most part, the random phase approximation, (see Chap. 8) is considered the appropriate theory and not the Tamm–Dancoff approximation, for the *ph* case. However, in order not to complicate things too much we shall restrict ourselves to the Tamm–Dancoff case.

The Tamm–Dancoff equation (see Chap. 8) has the following form.

$$\left(E_{\mu}^{\text{TDA}} - \epsilon_m + \epsilon_i\right) C_{mi}^{\mu} = \sum_{nj} \bar{v}_{mj,\,in} C_{nj}^{\mu}. \tag{4.91}$$

This equation, where m, n (i, j) are indices above (below) the Fermi level, is usually solved in a model space with, say, one shell below and one above the Fermi level. In analogy to the case of two valence nucleons, the phenomenological interaction entering Eq. (4.91) has to simulate the effect of the $1p$–$1h$ states not included in the model space, and also $2p$–$2h$, $3p$–$3h$, etc. effects.

In general, for a closed shell nucleus we can make the following expansion of such a vibrational state (see Chap. 8):

$$|\mu\rangle = \sum_{mi} C_{mi}^{\mu} a_m^{+} a_i |\text{HF}\rangle + \sum_{\substack{m<n \\ i<j}} C_{mn,\,ij}^{\mu} a_m^{+} a_n^{+} a_i a_j |\text{HF}\rangle + \cdots. \tag{4.92}$$

However, the procedure for arriving at an effective interaction starting from Eq. (4.92) has to be somewhat different from the valence nucleon case. First we introduce the projector P which projects onto the *whole* $1p$–$1h$ subspace, that is:

$$\begin{aligned} P &= \sum_{mi} a_m^{+} a_i |\text{HF}\rangle\langle\text{HF}| a_i^{+} a_m, \\ Q &= 1 - P, \end{aligned} \tag{4.93}$$

with no restriction on the sum. Using the same formalism as for arriving at Eq. (4.85), we obtain in the *ph* case

$$\left[E_\mu - E_0^{\mathrm{HF}} - \epsilon_m + \epsilon_i\right] C_{mi}^\mu = \sum_{nj} \langle mi | V_{\mathrm{eff}}^{E_\mu} | nj \rangle C_{nj}^\mu, \tag{4.94}$$

where V_{eff} is formally given by the same expression as in Eq. (4.83) with, however, the projection operators P and Q given by Eq. (4.93). The first contribution to Q comes from the $2p$–$2h$ components, that is, from Q_{2p-2h}. Calculating the contribution of this term to second order perturbation theory in V, we obtain:

$$\begin{aligned}
\left(V_{\mathrm{eff}\ mj,\ in}^{(2p-2h)}\right) = & \sum_{m'<n'} \bar{v}_{mj,\ m'n'} \frac{1}{\Omega_\mu + \epsilon_i + \epsilon_j - \epsilon_{n'} - \epsilon_{m'}} \bar{v}_{n'm',\ in} \\
& - \sum_{i'<j'} \bar{v}_{j'i',\ ni} \frac{1}{\Omega_\mu - \epsilon_m - \epsilon_n + \epsilon_{i'} + \epsilon_{j'}} \bar{v}_{jm,\ i'j'} \\
& - \sum_{n'j'} \bar{v}_{mj',\ nn'} \frac{1}{\Omega_\mu - \epsilon_n + \epsilon_i - \epsilon_{n'} + \epsilon_{j'}} \bar{v}_{jn',\ ij'} \\
& - \sum_{m'i'} \bar{v}_{ji',\ m'i} \frac{1}{\Omega_\mu - \epsilon_m + \epsilon_j - \epsilon_{m'} + \epsilon_{j'}} \bar{v}_{mm',\ i'n},
\end{aligned} \tag{4.95}$$

where $\Omega_\mu = E_\mu^A - E_0^{\mathrm{HF}}$ and terms which renormalize single-particle energies are not included. The first term on the r.h.s. of Eq. (4.95) together with the first order term of V_{eff} just gives the first two terms of the G-matrix:

$$G_{mj,\ in}^{\Omega_\mu + \epsilon_i + \epsilon_j}. \tag{4.96}$$

Indeed, we can verify that to each order in V there exists in V_{eff} a term which corresponds to the corresponding order in Eq. (4.96). Therefore, V_{eff} consists of a first term which is the G-matrix (4.96). The second term on the r.h.s. of Eq. (4.95) represents hole–hole correlations which numerically are found to be small and therefore may be neglected. The third and fourth term are analogous to the polarization diagram (Fig. 4.6) for the case of two valence nucleons. Again, these terms cannot be used as they stand, but we can show that it is a consistent re-summation of higher order terms to replace the bare matrix elements by their corresponding G-matrix elements. The polarization diagram in the *ph* case is displayed in Fig. 4.7. The fact that we already have a G-matrix in the full *ph* space to lowest order is the essential difference between the *ph* and *pp* cases. In recognizing this, we may now convert Eq. (4.94) to one in a model *ph* space. For this purpose we split the projector P into a part P_M which projects onto the model *ph* space and into a part P_R which projects onto the rest of the *ph* space:

$$P = P_M + P_R.$$

Eliminating the components in Eq. (4.94) that are outside the model space with the aid of the Feshbach projection operator formalism [Fe 62], which we have already used [see, for example, Eqs. (4.80) and (4.81)], leads to a TDA equation in the model space. The effective interaction appearing there can be expanded up to

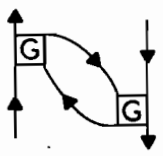


Figure 4.7. Polarization diagram in the particle-hole case.

second order in the G-matrix. The term which comes from Eq. (4.96) is shown in Fig. 4.8, where it should be appreciated that the summation over the intermediate lines goes only over states that are *not* in the modelspace. Realistic calculations [BK 69, KBB 70] show that the inclusion of the polarization diagram Fig. 4.7 has an important effect: Whereas RPA calculations (see Chap. 8) without it give imaginary solutions, the polarization contribution stabilizes the solutions. The reason for this—as discussed in more detail in Chap. 8—is that the RPA overestimates ground state correlations and polarization or screening terms have to be introduced to reduce them.

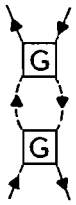


Figure 4.8. Second order contribution to V_{eff} in the model *ph* space. The broken intermediate lines shall indicate that one has to sum only over states not contained in the model space.

With these remarks we wish to finish our consideration of the microscopic effective interaction and, in the next section, turn to their phenomenological counterparts.

4.4 Phenomenological Effective Interactions

4.4.1 General Remarks

In Section 4.3 we saw how effective forces can be defined microscopically and how difficult it is in practice to calculate them and get quantitative agreement with experiment. Consequently, from the early days of nuclear physics the use of phenomenological forces, which contain a certain number of *fit parameters* adjusted to reproduce the experimental data has been adopted. In many cases this procedure has turned out to be extremely successful and, using only a few parameters chosen once and for all, much experimental data covering quite a large range of nuclear masses can be explained. Therefore, it is all the more disconcerting that a really satisfying microscopic theory able to explain the success of these phenomenological forces is somehow still missing.

There exists, of course, an enormous number of different phenomenological interactions that have been applied to problems in nuclear physics. Each of them has been used for a specific problem and their range of

validity and success varies very much. It lies outside the scope of this book to give a complete picture, and we will restrict our discussion to certain essential properties and the presentation of only the most successful examples. Most of them are only constructed for a special purpose, as, for example, for Hartree–Fock calculations (see Chap. 5), which calculate the nuclear self-consistent field, and bulk properties of nuclei such as binding energies and saturation densities. Others serve as effective forces between valence nucleons or between particles and holes (see Sec. 4.3). Therefore, we must be very careful in comparing these different types of interaction, even if, as we shall see, they look very similar in mathematical structure.

As we discussed in the last section, the effective interactions are used in a certain shell model configuration space: *the model space*. We therefore have to expect a certain dependence of the effective interaction on the space, that is, different interactions have to be taken for different sizes of the model space. As we have seen, the microscopically defined effective interaction is always energy dependent, while most of the time there is no explicit energy dependence in the phenomenological forces. It is usually sufficient to put all these dependences in a varying strength parameter.

It is evident that we prefer to choose the analytical form of the effective interaction to be as simple as possible. For example, it is often assumed that the effective interaction obeys the same *invariance principles* as the bare nucleon–nucleon interaction (see Sec. 4.2). This is certainly not always true. For instance, we should expect the renormalization procedure which describes the transition from the bare nucleon–nucleon interaction to the effective one to depend on the actual density of the system, that is, we get a different force in the interior of the system than in the surface and outer regions. For a shell model calculation in a fixed well, we should therefore not expect a translationally invariant residual interaction.

We know that the *range of the nuclear force* is rather short. We have seen in Section 4.3 that this is true even for the range of the effective G-matrix. The simplest ansatz therefore consists in using a *zero range* force whose radial dependence is described by a δ-function. In fact, such forces turn out to be rather useful because they are simple to handle and they describe many nuclear properties quite well. More realistic forces, however, need to have a finite non-vanishing range (see [AS 71, Sch 72b]). A *finite range* can be simulated by a *momentum dependence*. To show this we transform a function $V(\mathbf{r})$ of the relative distance $\mathbf{r}=\mathbf{r}_1-\mathbf{r}_2$ into momentum space

$$\langle \mathbf{p}|V|\mathbf{p}'\rangle = \frac{1}{(2\pi\hbar)^3}\int e^{-\frac{i}{\hbar}(\mathbf{p}-\mathbf{p}')\mathbf{r}} V(\mathbf{r}) d^3r. \tag{4.97}$$

We see that a δ-force is a constant and that any range represents a p-dependence in momentum space. The simplest rotationally invariant one is of the form

$$(2\pi\hbar)^3\langle \mathbf{p}|V|\mathbf{p}'\rangle = V_0 + V_1\mathbf{p}'^2 + V_1\mathbf{p}^2 + V_2\mathbf{p}\mathbf{p}', \tag{4.98}$$

which in coordinate space corresponds to the momentum dependent operator

$$V(\mathbf{r}) = V_0\delta(\mathbf{r}) + V_1\left(\hat{\mathbf{p}}^2\delta(\mathbf{r}) + \delta(\mathbf{r})\hat{\mathbf{p}}^2\right) + V_2\hat{\mathbf{p}}\delta(\mathbf{r})\hat{\mathbf{p}}. \tag{4.99}$$

Effective forces usually depend on the density $\rho(\mathbf{r})$. Such *density dependence* is easy to understand if, for example, we consider expression (4.39) for the G-matrix: The range of the summation depends on the Fermi energy, which is itself a function of the density. It is also quite easy to see that the different projection operators Q entering the definition of the effective interactions [see Eqs. (4.40) and (4.77)] can be expressed as a nonlocal density which is therefore another source of density dependence.

Other simple forms that work surprisingly well are separable two-body forces, that is, forces which can be represented as products (or as a sum of products), in which each factor acts on only one of the two particles.

4.4.2 Simple Central Forces

There exists a large number of forces which use only the central part (4.20), but differ in their radial dependence $V(r)$. Some simple potentials that have been used are, for example, the Yukawa potential (4.33) or

$$V(r) = -V_0 e^{-r^2/r_0^2} \qquad \text{(Gauss potential)}, \tag{4.100}$$

$$V(r) = -V_0 \frac{e^{-r/r_0}}{1-e^{-r/r_0}} \qquad \text{(Hulthen potential)}, \tag{4.101}$$

$$V(r) = -V_0\delta\left(\frac{\mathbf{r}}{r_0}\right) \qquad \text{(contact potential)}. \tag{4.102}$$

The potential depth V_0 and the range r_0 are adjusted to fit experimental data. One finds $V_0 \simeq 50$ MeV and $r_0 \simeq 1-2$ fm (the Compton wavelength of the pion is 1.4 fm).

For the r-dependent coefficients $V(r)$ in Eq. (4.20) the same radial dependence is usually assumed; only their strength and sign are adjusted to fit empirical data. Here we do not wish to discuss which one of the

Table 4.3 Coefficients for the exchange mixtures

Mixture	a_{et}	a_{es}	a_{ot}	a_{os}
Wigner	1	1	1	1
Kurath [Ku 56, SW 66]	1	0.6	−0.6	−1
Serber [HS 57]	1	1	0	0
Rosenfeld [EF 57]	1	0.6	−0.34	−1.78
Ferrel–Visscher [VF 57]	1	0.634	−0.366	0
Soper [So 57]	1	0.46	0.14	−0.4
Gillet [GGS 66] I	1	0.6	−0.6	0.6
II	1	0.4	−0.2	0.2

different exchange mixtures is to be preferred; we wish only to give some of the most commonly used values of the coefficients (Table 4.3).

$$V(1,2)=V_0 f\left(\frac{r}{r_0}\right)(a_{et}\Pi_e^r\Pi_t^\sigma+a_{es}\Pi_e^r\Pi_s^\sigma+a_{ot}\Pi_o^r\Pi_t^\sigma+a_{os}\Pi_o^r\Pi_s^\sigma). \tag{4.103}$$

Other forces of a similar type have been used by Kallio and Koltveit [KK 64] and by Elliott and Clark [CE 65]. The latter authors also include a tensor term and a two-body spin orbit force.

4.4.3 The Skyrme Interaction

In 1956 Skyrme [Sk 56, 59] proposed an effective interaction with a three-body term viz:

$$V=\sum_{i<j}V(i,j)+\sum_{i<j<k}V(i,j,k). \tag{4.104}$$

To simplify the calculations, he used a short-range expansion in the form of Eq. (4.99) for the two-body part:

$$\begin{aligned}V(1,2)=&\,t_0(1+x_0P^\sigma)\delta(\mathbf{r}_1-\mathbf{r}_2)\\&+\tfrac{1}{2}t_1\left[\delta(\mathbf{r}_1-\mathbf{r}_2)\mathbf{k}^2+\mathbf{k}^2\delta(\mathbf{r}_1-\mathbf{r}_2)\right]+t_2\mathbf{k}\delta(\mathbf{r}_1-\mathbf{r}_2)\mathbf{k}\\&+iW_0(\boldsymbol{\sigma}^{(1)}+\boldsymbol{\sigma}^{(2)})\mathbf{k}\times\delta(\mathbf{r}_1-\mathbf{r}_2)\mathbf{k},\end{aligned} \tag{4.105}$$

where $\mathbf{k}=(1/\hbar)\mathbf{p}$ is the operator of the relative momentum

$$\mathbf{k}=\frac{1}{2i}(\nabla_1-\nabla_2). \tag{4.106}$$

For the three-body force Skyrme also assumed a zero range force of the form

$$V(1,2,3)=t_3\delta(\mathbf{r}_1-\mathbf{r}_2)\delta(\mathbf{r}_2-\mathbf{r}_3). \tag{4.107}$$

The five constants—t_0, t_1, t_2, t_3, x_0—and W_0 were adjusted to the experimental binding energies and radii. There are several sets of parameters called Skyrme I, II, etc. (see Sec. 5.6) resulting from different fits. We present here Skyrme III [BFG 75]:

$$\begin{aligned}t_0&=-1128.75\text{ MeV fm}^3; & t_1&=395.0\text{ MeV fm}^5;\\ t_2&=-95.0\text{ MeV fm}^5; & t_3&=14000.0\text{ MeV fm}^6;\\ W_0&=120\text{ MeV fm}^5; & x_0&=0.45.\end{aligned} \tag{4.108}$$

The parameter t_0 describes a pure δ-force with a spin-exchange; t_1 and t_2 simulate an effective range, as in Eq. (4.99). The fourth term in Eq. (4.105) represents a two-body spin orbit interaction. It can be obtained [BS 56] from a normal spin orbit term [see Eq. (4.23)] in the short range limit.

In Chapter 5, we shall see that this force has been used extensively in

Hartree–Fock calculations. For spin saturated even–even nuclei, the three-body term (4.104) turns out to be equivalent to a density dependent two-body interaction*:

$$V_\rho(1,2)=\frac{1}{6}t_3(1+P^\sigma)\delta(\mathbf{r}_1-\mathbf{r}_2)\rho(\tfrac{1}{2}(\mathbf{r}_1+\mathbf{r}_2)). \tag{4.109}$$

Such a density dependent term can also be regarded as the phenomenological representation of the ρ-dependence of the microscopic effective interaction. This interpretation is preferrable to the view that the Skyrme force contains a three-body interaction, since we know that three-body interactions are rather weak in nuclei.

There are essentially three reasons why this force has gained so much importance over the last ten years:

(i) Vautherin and Brink [VB 72] (see Chap. 5) were able to reproduce the nuclear binding energies over the whole periodic table with a reasonable set of parameters and, at the same time, the nuclear radii. This had not been possible with the usual non-density dependent forces.

(ii) Negele and Vautherin [NV 72] gave the connection between this force and the more fundamental G-matrix discussed in the last sections.

(iii) The mathematical form of the force is extremely simple. The δ-functions simplify all types of calculations enormously.

There are similar types of interactions originally proposed by Moszkowski [Mo 70], the so-called *modified δ-interactions* (MDI). They differ from the Skyrme interaction by the absence of the t_2-term and the spin orbit force and also by a different ρ-dependence ($\sim\rho^\sigma$ with $\sigma<1$).†

4.4.4 The Gogny Interaction

Despite the great success of the Skyrme interaction, it has been argued that zero range forces might not be able to simulate the long range or even the intermediate range parts of the realistic effective interaction. In particular, the present versions of the Skyrme force are not able to properly describe pairing correlations in nuclei (see Chap. 6), therefore Gogny [Go 75b] replaced the t_0, t_1 and t_2 parts of the Skyrme force by a sum of two

*This is no longer true for systems without spin saturation. In such cases, the three-body term (4.107) in the Skyrme force, which is purely local and repulsive, favors parallel spin alignment, that is, nuclear ferromagnetism. This contradicts the observed spin saturation and the pairing properties in nuclei. This difficulty can be avoided by using either a density dependent two-body interaction [BJS 75] of the form (4.109) or a nonlocality in the three-body term [ON 78].

†There is a strong correlation between the nuclear compressibility and the exponent σ in the density dependence [CS 72].

Gaussians* with spin–isospin exchange mixtures (a force which was originally used by Brink and Booker [BB 67]) and got

$$V(1,2)=\sum_{i=1}^{2} e^{-(\mathbf{r}_1-\mathbf{r}_2)^2/\mu_i^2}(W_i+B_iP^\sigma-H_iP^\tau-M_iP^\sigma P^\tau)$$
$$+iW_0(\boldsymbol{\sigma}_1+\boldsymbol{\sigma}_2)\mathbf{k}\times\delta(\mathbf{r}_1-\mathbf{r}_2)\mathbf{k}$$
$$+t_3(1+P^\sigma)\delta(\mathbf{r}_1-\mathbf{r}_2)\rho^{1/3}(\tfrac{1}{2}(\mathbf{r}_1+\mathbf{r}_2)). \qquad (4.110)$$

The parameters were adjusted to the properties of finite nuclei, and for nuclear matter (Table 4.4).

Table 4.4 Force parameters of the Gogny force (D1)

i	μ_i [fm]	W_i	B_i	H_i	M_i [MeV]	$W_0=+115$ [MeV fm^5]
1	0.7	−402.4	−100.	−496.2	−23.56	$t_3=1350$ [MeV fm^4]
2	1.2	−21.30	−11.77	37.27	−68.81	

4.4.5 The Migdal Force

This force was proposed by Migdal [Mi 67] in his theory of finite Fermi systems. Based on the interacting quasi-particle concept of Landau's theory of a Fermi liquid [La 59], Migdal introduces this force to describe the collective excitations in nuclei.†

Starting from the ground state of an even–even system, the quasi-particles are defined as the low-lying excitations in the neighboring odd mass nuclei. The ground state of the even system contains no quasi-particles, and excited states are characterized by the quasi-particle occupation numbers n_λ. A change of these occupation numbers by the amount δn_λ causes a change in the total energy E_0 of the system by the amount

$$\delta E_0=\sum_\lambda \epsilon_\lambda^0\delta n_\lambda+\frac{1}{2}\sum_{\lambda\lambda'}F_{\lambda\lambda'}\delta n_\lambda\delta n_{\lambda'},$$

where ϵ_λ^0 are the energies of a quasi-particle λ in the absence of any other quasiparticle and $F_{\lambda\lambda'}$ is the so-called quasi-particle interaction. Migdal introduces an effective particle–hole interaction F^ω and an effective particle–particle (or hole–hole) interaction F^ξ.

In an infinite system with translational invariance, the quasi-particles are characterized by the momentum $\mathbf{k}$, and Landau could show that the *ph*-interaction $F(\mathbf{k},\mathbf{k}')$ is given by the second derivative of the total energy

*Similar density dependent effective interactions have been used in [Kr 70, ZR 71, RPS 72, LMV 73, RBP 77].

†It would be beyond the scope of this book to go into the details of this theory. The reader may, however, find some basic ingredients in Section F.7.

E_0 with respect to the quasi-particle densities $n(\mathbf{k})$:

$$F^{ph}(\mathbf{k},\mathbf{k}')=\frac{\delta^2 E_0}{\delta n(\mathbf{k})\delta n(\mathbf{k}')}. \tag{4.111}$$

At the Fermi surface this is an exact relation.

In a finite system the quasi-particle density $\bar{\rho}_{\lambda\lambda'}$ is no longer completely determined by its diagonal elements (the occupation numbers n_λ), but it also contains information about the form of the single-particle wave functions φ_λ. The effective interaction then depends on four indices, and it has been proposed to derive this quantity, in analogy to Eq. (4.111), from the exact ground state energy [Br 71]

$$F^{ph}_{psqr}=\frac{\delta E_0}{\delta\bar{\rho}_{qp}\delta\bar{\rho}_{rs}}. \tag{4.112}$$

So far it has not been shown that this is an exact relation. However, a very similar expression is obtained in a quite different (*approximate*) theory: the time dependent Hartree–Fock theory in the limit of a motion with small amplitudes (see Sec. 8.5). Starting from the assumption that the wave function is a Slater determinant (i.e., $\rho^2=\rho$; see Sec. 5.3.3) and that the total energy can be expressed by a functional $E_0[\rho]$, we obtain in this case the effective *ph*-interaction as the second derivative of $E_0[\rho]$ with respect to ρ [Eq. (8.124)], just as in Eq. (4.112).

It is clear from the above that, for instance, the Skyrme force (Sec. 4.4.3) cannot be compared with the Migdal force directly, even though, as we shall see, it looks very similar. Within the approximation $\rho^2\simeq\rho$, however, an indirect relation may be established by differentiating the ground state energy calculated with the Skyrme interaction twice with respect to the density.

Like the Skyrme force, the Migdal force is an expansion in momentum space. However, contrary to potentials suitable for Hartree–Fock calculations (such as the Skyrme force), the p^2-terms do not play an essential role, as the Migdal forces do not have to guarantee saturation (they are constructed to describe different physical situations anyway). In most calculations it is therefore sufficient to take into account only the constant in momentum space which gives a pure δ-force in coordinate space. On the other hand, spin and isospin exchange mixtures are now very important. They are different for the particle–particle and particle–hole forces:

$$V(1,2)=V_0\delta(\mathbf{r}_1-\mathbf{r}_2)(f+f'\boldsymbol{\tau}^{(1)}\boldsymbol{\tau}^{(2)}+g\boldsymbol{\sigma}^{(1)}\boldsymbol{\sigma}^{(2)}+g'\boldsymbol{\sigma}^{(1)}\boldsymbol{\sigma}^{(2)}\boldsymbol{\tau}^{(1)}\boldsymbol{\tau}^{(2)}). \tag{4.113}$$

V_0 is a strength parameter which has to be adjusted to the configuration space (e.g., in the ^{208}Pb region $V_0=380$ MeV fm^3). Guman and Birbrair [GB 65] proposed to take into account the different interaction strengths inside and outside the nucleus and the diffuseness of the nuclear surface by a linear density dependence of the constants f, f',

$$f=f^{ex}+(f^{in}-f^{ex})\rho(r), \tag{4.114}$$

where $\rho(r)$ has the form of a Fermi distribution

$$\rho(r)=\frac{1}{1+\exp[(r-R)/a]}. \tag{4.115}$$

The additional parameters R and a represent the radius and the diffuseness, respectively, of the nucleus. Contrary to a Hartree–Fock calculation with a density dependent interaction (see Chap. 5), the density (4.114) is not adjusted self-consistently. Therefore, the Migdal force violates translational invariance. Of course, this is no drawback, since the renormalization procedure is closely connected with the underlying single-particle potential, which also violates translational invariance. In fact, a proper choice of the effective residual interaction should restore this invariance (see Mikeska and Brenig [MB 69]). From this condition one can deduce additional relations among the parameters f, f', g, g' [NW 72, 74].

The Migdal force has been widely used to calculate low-lying collective vibrations in nuclei within the framework of the random phase approximation (see Chap. 8). The effective charges caused by such vibrations (see Chap. 9) provide an enormous amount of experimental data with which to adjust the six parameters f^{in}, f^{ex}, f'^{in}, f'^{ex}, and g, g' for the particle–hole and particle–particle forces (for details, see [RBS 73, BSK 73, BER 75]). For the particle–hole force Ring and Speth [RS 74a] found:

$$\begin{aligned} &f^{in}=0.0685; \quad f'^{in}=0.3315; \quad f^{ex}=-2.165; \\ &f'^{ex}=0.465; \quad g=0.575; \quad g'=0.725. \end{aligned} \tag{4.116}$$

It is important to note that this is an effective *ph*-force which does not have to be antisymmetrized [KMS 76]. It shows strong attraction outside the nucleus and is close to zero inside the nucleus.

4.4.6 The Surface-Delta Interaction (SDI)

Like the Migdal interaction, this force is thought of as an effective residual interaction among the particles near the Fermi surface. The main physical idea is that the nucleons move almost independently in the nuclear interior. In fact, the residual interaction of Migdal is rather weak inside the nucleus. Most of the collisions occur at the nuclear surface where the Pauli principle loses its importance and the nucleons feel the strong attractive interaction [GM 65]. The behavior of the force outside the surface is again not very important because there the wave functions have exponentially decaying tails, that is, the probability of finding a nucleon there goes rapidly to zero. Therefore, it is a meaningful approximation to restrict the whole interaction to the nuclear surface and to define the so-called surface delta interaction:

$$V(1,2)=-V_0\delta(\mathbf{r}_1-\mathbf{r}_2)\delta(|\mathbf{r}_1|-R_0). \tag{4.117}$$

V_0 can still depend on the spin and isospin coordinates in the usual way

(4.20). This force has very simple geometrical properties. Using the expansion of the δ-function in Legendre polynomials—

$$\delta(\mathbf{r}_1-\mathbf{r}_2)=\frac{\delta(r_1-r_2)}{r_1 r_2}\sum_l \frac{2l+1}{4\pi}P_l(\cos\theta_{12}), \tag{4.118}$$

where θ_{12} is the angle between the vectors $\mathbf{r}_1$ and $\mathbf{r}_2$—and the addition theorem of the spherical harmonics [Ed 57, Eq. 4.6.6], we get:

$$V(1,2)=-V_0\sum_{lm}\frac{\delta(r_1-R_0)}{r_1}Y^*_{lm}(\theta_1\varphi_1)\frac{\delta(r_2-R_0)}{r_2}Y_{lm}(\theta_2\varphi_2). \tag{4.119}$$

This is an infinite sum over separable terms. In a spherical basis, the matrix elements coupled to a good angular momentum become extremely simple because only one term of the sum in (4.119) survives. This model force has been applied [KS 63, PAM 66, FP 67, Fä 68, VPK 69] in many calculations. It has been used—like the Migdal force—as an effective force among the valence particles. It is, however, not meaningful to extend the underlying configuration space over more than two major shells, because in that case we have levels with the same angular momentum quantum numbers. Since the radial integrals are approximated by the value of the wave function at $r=R_0$, there is no cancellation for wave functions with different numbers of radial nodes. Consequently, the particles in such levels feel an unphysical strong interaction.

4.4.7 Separable Forces and Multipole Expansions

In the last section we saw that the surface-delta interaction is extremely simple to handle because of its separability in a shell model basis.

We call a force *separable in particle hole direction* if it can be written in second quantization in the form [: : means normal ordering in the sense of Eq. (C.50)]

$$V=:Q^+\cdot Q:, \tag{4.120}$$

where

$$Q=\sum_{k_1k_2}q_{k_1k_2}a^+_{k_1}a_{k_2} \tag{4.121}$$

is a one-particle operator. Acting on a Slater determinant, it creates a superposition of *ph*-pairs. A force is called *separable in pp-direction* if it has the form

$$V=P^+\cdot P, \tag{4.122}$$

where

$$P^+=\frac{1}{2}\sum_{k_1k_2}p_{k_1k_2}a^+_{k_1}a^+_{k_2} \tag{4.123}$$

is a pair creation operator. Acting on a Slater determinant, it creates a

superposition of *pp*-pairs. It seems very unlikely that a realistic force can be represented by so simple an ansatz as Eqs. (4.120) and (4.122), however, it has been shown that the matrix elements of these schematic forces with a suitably chosen strength are of the same magnitude as those calculated from a microscopic G-matrix with polarization terms (see Sec. 4.3 and F.5) [BK 67b].

We wish here to study the question of *separability* somewhat further and investigate a general *phenomenological effective force* without spin and isospin dependence. The force shall depend only on the relative distance between the nucleons, therefore we can expand in the following way [La 64].

$$V(|\mathbf{r}_1-\mathbf{r}_2|)=\sum_l V_l(r_1,r_2)\sum_m Y^*_{lm}(\theta_1\varphi_1)\,Y_{lm}(\theta_2\varphi_2), \tag{4.124}$$

with

$$V_l(r_1,r_2)=2\pi\int_{-1}^{1} V(|\mathbf{r}_1-\mathbf{r}_2|)P_l(\cos\theta_{12})\,d\cos\theta_{12}. \tag{4.125}$$

For a δ-force, $V=-V_0\delta(\mathbf{r})$, we find that V_l does not depend on l:

$$V_l(r_1,r_2)=-V_0\frac{\delta(r_1-r_2)}{r_1r_2}. \tag{4.126}$$

In Eq. (4.124) we have written a general force as an infinite sum of terms, each of which are separable in the angular coordinates. As we see from the δ-force, we cannot expect this expansion to converge rapidly for a short range force. In fact, we obtain an estimate for the effective range of each component in the expansion (4.124) if we restrict ourselves to a small region in space where $r_1\simeq r_2\simeq r$. The function $P_l(\cos\theta_{12})$ has its major contributions only in the region $0\leqslant\theta_{12}\lesssim 1/l$ (Fig. 4.9), that is, V_l (4.125) is small, if the range of the potential $V(r_{12})$ is large compared to r/l, that is,

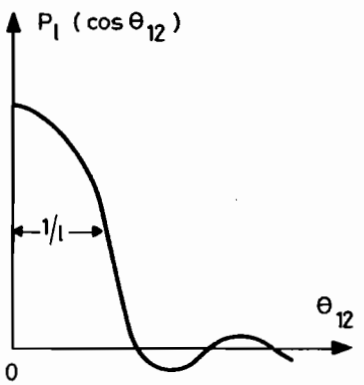


Figure 4.9. Legendre polynomial of degree l as a function of the angle θ_{12} between $\mathbf{r}_1$ and $\mathbf{r}_2$.

the larger the strength of the high l-components V_l, the shorter the range of the force. In a force with zero range (δ-force), all l-components have the same relative weight.

We now come back to the question of separability. The general force (4.124) and the δ-force (4.126) are not separable because of the radial dependence in $V_l(r_1, r_2)$. On the other hand, if one assumes that the interaction is surface peaked as the SDI-interaction (4.117), it then becomes a sum of separable terms because of the additional factor $\delta(r-R_0)$.

$$V_l^{\mathrm{SDI}}(r_1 r_2) = \frac{-V_0}{r_1}\delta(r_1 - R_0)\frac{1}{r_2}\delta(r_2 - R_0). \tag{4.127}$$

A more general separable ansatz of this kind is

$$V_l(r_1 r_2) = f_l(r_1)\cdot f_l(r_2). \tag{4.128}$$

One often chooses $f_l \sim r^l$ and obtains the multipole–multipole forces

$$V = -\frac{1}{2}\sum_{lm}\chi_l : Q_{lm}^{+} Q_{lm} : = -\frac{1}{2}\sum_{l}\chi_l : Q_l^{+}\cdot Q_l :, \tag{4.129}$$

where χ_l are constants and Q_{lm} is the multipole operator

$$Q_{lm} = \sum_{k_1 k_2}\langle k_1 | r^l Y_{lm}(\theta, \varphi) | k_2\rangle a_{k_1}^{+} a_{k_2}. \tag{4.130}$$

If we take into account only the components with small l-values, we obtain long range forces. The most well known is the quadruple–quadruple force [El 58, KS 63]. We shall see in Chapter 7 that it can be used to describe the quadrupole deformations of nuclei self-consistently.

Again we must emphasize that these multipole forces are only to be considered as an effective force between valence particles within a restricted configuration space. If there are many such valence particles we can apply a Hartree–Fock approximation for them* (see Chap. 5). However, such HF-calculations with multipole forces for valence nucleons should not be mixed up with the basic HF-calculations for the binding energies.

The expansion (4.124) turns out to be very useful for the *long range part* of the effective interaction, because in this case the functions V_l become small for higher l-values. To obtain a similar description of the *short range part* one uses a different kind of expansion. We shall restrict outselves to interactions between nucleons of the same type (for the treatment of short range correlations between protons and neutrons, see Sec. 7.5).

We now consider the general expression (4.3) of a not necessarily local force $V(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_1', \mathbf{r}_2')$. In the case of a local force, we have $\mathbf{r}_1 = \mathbf{r}_1'$ and $\mathbf{r}_2 = \mathbf{r}_2'$, and the expansion (4.124) is made in the variables $\mathbf{r}_1, \mathbf{r}_2$. Now we require

*In such calculations, the exchange term (5.43) to the HF potential is usually neglected because it is small [BK 68, Ma 74].

instead $\mathbf{r}_1=\mathbf{r}_2$ and $\mathbf{r}'_1=\mathbf{r}'_2$. We expand in $\mathbf{r}_1, \mathbf{r}'_1$ and obtain expressions similar to (4.124) and (4.125).

$$\delta(\mathbf{r}_1-\mathbf{r}_2)\delta(\mathbf{r}'_1-\mathbf{r}'_2)V(\mathbf{r}_1,\mathbf{r}'_1) = \delta(\mathbf{r}_1-\mathbf{r}_2)\delta(\mathbf{r}'_1-\mathbf{r}'_2)\sum_{lm} V_l(r_1,r'_1)\,Y_{lm}(\theta_1\varphi_1)Y^*_{lm}(\theta'_1\varphi'_1). \quad (4.131)$$

The requirements $\mathbf{r}_1=\mathbf{r}_2$ and $\mathbf{r}'_1=\mathbf{r}'_2$ already restrict us to a short range because it is only true if the two nucleons are at the same place. (This does not necessarily mean a δ-force, because we are allowing for nonlocal forces). At a first glance, the expansion (4.131) seems to be as poor as Eq. (4.124) for a short range force. However, we have to calculate matrix elements between two-particle wave functions. As we shall demonstrate in a moment, the lth term in (4.131) gives a contribution only to the matrix element when the two particles are coupled to spin $I=l$. For two particles in a single j-shell, for instance, we find that their wave functions have maximal spatial overlap for $I=0$. In this case, the angular part of their wave function takes the form:

$$\langle\Omega_1\Omega_2 \mid jjI=0, M=0\rangle \propto \sum_{m=-l}^{l}(-)^m Y_{lm}(\Omega_1)\,Y_{l-m}(\Omega_2) \propto P_l(\cos\theta_{12}). \quad (4.132)$$

(Because of the short range force, we assume only the spin singlet to be important.) Therefore, we find maximal interaction for $I=0$, where the angular momenta $\mathbf{j}_1$ and $\mathbf{j}_2$ are antiparallel, and we can thus understand that the expansion (4.131) is meaningful for short range interactions.

Again, if we replace the non-separable radial term in (4.131) by a separable ansatz as in (4.129), we obtain

$$V_l(r_1,r'_1) = -G_l r_1^l \cdot r_1'^l, \quad (4.133)$$

$$V = -\sum_{lm} G_l P^+_{lm} P_{lm} = -\sum G_l P_l^+ \cdot P_l, \quad (4.134)$$

with

$$P^+_{lm} = \tfrac{1}{2}\sum_{k_1k_2}\langle k_1|r^l Y_{lm}|k_2\rangle a^+_{k_1}a^+_{\bar{k}_2}, \quad (4.135)$$

where $|\bar{k}\rangle$ is the time-reversed state to $|k\rangle$. In deriving Eq. (4.134) from (4.131) we must take care in handling the spin coordinate s, because the operator (4.133) is not separable in spin space a priori. Because of the δ-force character, however, we always have $s_1=-s_2$, $s'_1=-s'_2$, which makes it possible to derive (4.134).

We are now able to calculate the matrix elements of a δ-force in a single j-shell, where the single-particle wave functions are given by $\langle\mathbf{r}|nljm\rangle = R_{nl}(r)\langle\Omega|ljm\rangle$. We construct pair operators A^+_{IM} with angular momentum I, M:

$$A^+_{IM} = \frac{1}{\sqrt{2}}\sum_{m_1m_2} C^{j\ j\ I}_{m_1m_2M}\,a^+_{jm_1}a^+_{jm_2} \quad (4.136)$$

and use Eqs. (4.124) and (4.126) to obtain

$$V_\delta = -\sum_{I=0,2,\ldots} V_\delta(I) \sum_M A_{IM}^+ A_{IM},$$

with*

$$V_\delta(I) = V_0 \cdot R_4 \frac{1}{2} \frac{1}{2I+1} |\langle j \| Y_I \| j \rangle|^2 = \frac{V_0}{8\pi} R_4 \frac{(2j+1)^2}{2I+1} \left| C^{j\ j\ I}_{1/2\ -1/2\ 0} \right|^2, \tag{4.137}$$

where R_4 is the radial integral

$$R_4 = \int_0^\infty dr\, r^2 R_{nl}^4(r).$$

From (4.137) we see that only the lth term in the expansion (4.131) goes into the matrix element coupled to spin $I = l$. In the case of a δ-force, $V_\delta(I)$ is maximal for $I = 0$ and drops off rapidly with increasing I (Table 4.5).* The most important part of the force (4.134) is therefore the $l = I = 0$ part.

Table 4.5 The relative magnitude of the matrix elements $V_\delta(I)$ (4.137) in a single j-shell

$V_\delta(I)/V_\delta(0)$	$I=0$	2	4	6	8	10	12
$j=7/2$	1	0.238	0.117	0.058	—	—	—
9/2	1	0.242	0.126	0.075	0.040	—	—
11/2	1	0.245	0.131	0.082	0.053	0.025	—
13/2	1	0.246	0.133	0.087	0.060	0.038	0.019

To avoid complicated formulas, one usually introduces a slight modification of the Condon–Shortley phases (see [Ed 57]), the so-called BCS phases, and defines for $m > 0$:

$$|nljm\rangle_{\text{BCS}} = |nljm\rangle_{\text{CS}},$$
$$|nlj-m\rangle_{\text{BCS}} = (-)^{l+j-m} |nlj-m\rangle_{\text{CS}}. \tag{4.138}$$

The time reversal operator T [Eq. (2.45)] has a very simple form in this case. For $m > 0$, we find

$$|\overline{nljm}\rangle = T|nljm\rangle = |nlj-m\rangle; \qquad T|nlj-m\rangle = -|nljm\rangle. \tag{4.139}$$

The $l = 0$ part of Eq. (4.134), the so-called *pure pairing force*, has the form

$$V^p = -G\frac{1}{4} \sum_{\substack{kk' \\ \lesssim 0}} a_{k'}^+ a_{\bar{k}'}^+ a_{\bar{k}} a_k, \tag{4.140}$$

where $\bar{k}$ is again the time reversed state to k and the sum runs over all

* In the limit of large I- and j-values we find, by a semiclassical expansion [MJB 75],

$$\left| C^{j\ j\ I}_{1/2\ -1/2\ 0} \right|^2 \simeq \frac{2}{\pi \cdot j} \sqrt{1 - \left(\frac{I}{2j}\right)^2}\ .$$

values of k, k'. This force has been widely used to describe short range correlations in nuclei within the BCS-formalism (see Chap. 6). It is again important to notice that it should only be applied in a restricted configuration space. The value of the force constant is connected with the size of this space. It is easy to generalize this simple model force and to take into account higher l-values in the expansion (4.131). They are called *multipole pairing forces*. They are separable in particle–particle direction and play an important role in the description of pairing vibrations (see Chap. 8 and [BB 71, BBN 74b]).

A very simple model force for heavy nuclei has been obtained by combining a quadrupole–quadrupole force and a pure pairing force to the pairing-plus-quadrupole force [KS 63, BK 68]:

$$V^{QP} = -\tfrac{1}{2}\chi : Q_2^+ \cdot Q_2 : - G P_0^+ \cdot P_0, \tag{4.141}$$

which takes into account the most important long-range correlations and the most important short-range correlations, and is very easy to handle (see Sec. 7.4). It contains basically three constants: χ and two different constants G_p and G_n for protons and neutrons. It should only be applied in a configuration not larger than one major shell. For actual values of χ and G, see [KS 60] or [BK 68].

4.4.8 Experimentally Determined Effective Interactions

Since the derivation of the effective interaction from a bare nucleon–nucleon force is rather complicated, some authors have tried to determine the matrix elements of the nucleon–nucleon force in a certain basis directly from the experimental data (for a review, see [ST 76] and [BPO 76]).

Elliott et al. [EJM 68] used the experimental phase shifts of nucleon–nucleon scattering to derive the matrix elements of an effective interaction directly in an oscillator basis (*Sussex force*). In this way, the problem of the hard core is avoided completely.

Less ambitious are attempts to derive special matrix elements from the spectra of "simple" nuclei. These are nuclei in which one has good reason to believe that their structure is determined only by a few matrix elements whose number does not exceed the number of the observed levels [Ta 62]. We may take, for example, the $1f7/2$ shell nuclei, where the $f7/2$ shell is well separated in the simple particle spectrum. One assumes in this case that there exist many pure $f7/2$ configurations in nuclei close to the double magic core ^{40}Ca with $N \geqslant 20$ and $Z \geqslant 20$. These levels are completely determined by the eight matrix elements $\langle (f7/2)^2 IT | V | (f7/2)^2 IT \rangle$ for $I = 0, 1, \ldots, 7$ ($T = 1$ for I even, $T = 0$ for I odd).

McCullen et al. [MBZ 64] have determined these matrix elements from the spectrum of the nucleus ^{42}Sc. It contains one neutron and one proton outside the ^{40}Ca core. Coupling to angular momentum I gives the interaction energy

$$\begin{aligned} V(I) &= \langle j_1 j_2 I | V | j_1 j_2 I \rangle \\ &= B(I, {}^{42}\mathrm{Sc}) - B(I, {}^{41}\mathrm{Sc}) - B(I, {}^{41}\mathrm{Ca}) + B(I, {}^{40}\mathrm{Ca}), \end{aligned} \tag{4.142}$$

where B are the experimental binding energies of the level with spin I in ^{42}Sc and of the ground state of the other nuclei. Using these matrix elements, many energy levels and electromagnetic properties of the other $f7/2$ nuclei with more valence nucleons can be reproduced quite well. However, levels with a more complicated structure (e.g., holes in the ^{40}Ca core) are also encountered and it is not known how much they mix with the pure $f7/2$ configurations.

Schiffer [Sch 71b, 72b] and Molinari et al. [MJB 75] have made a more general investigation of many such simple nuclei with pure configurations where one nucleon is in a shell j_1 and the other in a shell j_2. The coupling of angular momentum yields multiplets with $|j_1-j_2| \leqslant I \leqslant j_1+j_2$. The interaction produces a splitting of the multiplets. It is convenient to plot not $V(I)$ (Eq. 4.137) versus I but rather the dimensionless ratio $V(I)/\bar{V}$ as a function of the "overlap angle" θ_{12} where $\bar{V}$ is the average two-body interaction energy

$$\bar{V} = \sum_I (2I+1)V(I) / \sum_I (2I+1). \tag{4.143}$$

This angle is the angle between the classical orbits of the two valence nucleons

$$\cos\theta_{12} = \frac{\mathbf{j}_1 \cdot \mathbf{j}_2}{|j_1|\,|j_2|} = \frac{I(I+1)-j_1(j_1+1)-j_2(j_2+1)}{2[j_1(j_1+1)]^{1/2}\cdot[j_2(j_2+1)]^{1/2}} \tag{4.144}$$

and it measures the spacial overlap of the two wave functions. In this plot (Fig. 4.10), the points for many different nuclei lie nearly on the same curves. As an example we show the plots for nuclei where the two valence nucleons are in the same orbit (Fig. 4.10) and compare them with the corresponding values for a pure δ-force. For identical particles ($T=1$) we again find that the absolute interaction

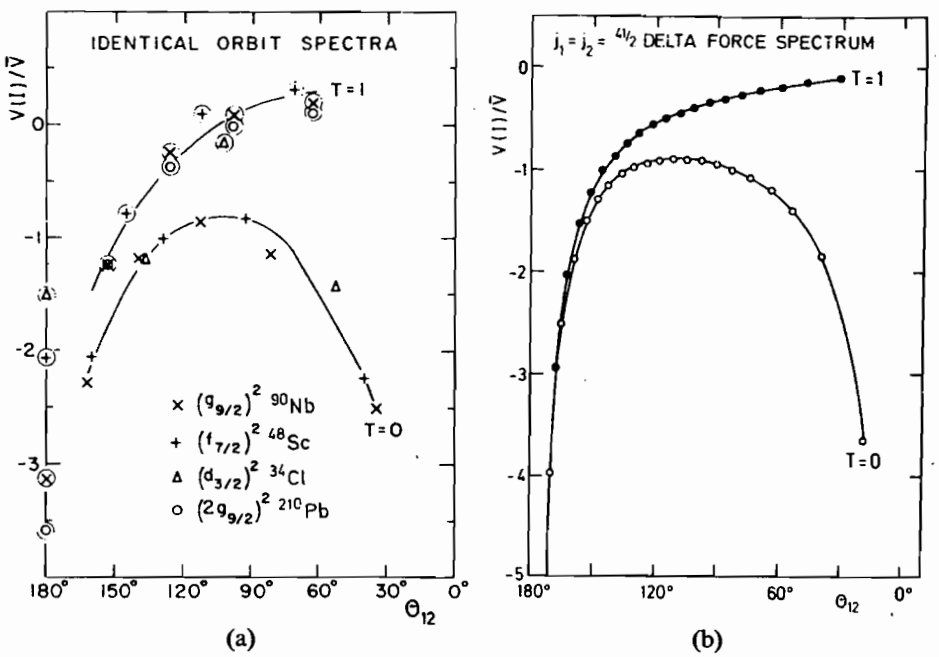


Figure 4.10. The relative interaction matrix elements $V(I)/\bar{V}$ as a function of the overlap angle θ_{12}. Experimental values for (a) four nuclei, (b) a pure δ-force spectrum and $j=41/2$. (From [AS 71].)

matrix elements are strongest for $\theta_{12}=180°$, that is, for antiparallel spins ($I=0$) and they drop off rapidly. For nonidentical particles in the case $T=0$ they also raise for $\theta_{12}=0$, that is, for parallel spins. For identical particles this configuration is prevented by the Pauli principle. This expresses the fact that a short-range force requires a large spacial overlap of the wave functions and it seems that an effective interaction of the δ-type can reproduce qualitatively the "experimentally" determined matrix elements. However, a quantitative comparison [Sch 72b, GSZ 74, AEE 78] shows that a long-range part is also needed to reproduce properties of odd–odd nuclei. The origin of this long-range part is as yet not completely understood. For a discussion of this point see [Mo 76c].

4.5 Concluding Remarks

In the second section of this chapter we discussed the bare forces acting between nucleons. We first restricted their analytical form, observing that they have to obey a certain number of invariance principles. It is well known that the long-range part of the nucleon–nucleon interaction is given by the one-pion exchange; at shorter distances, exchanges of two pions and heavier mesons become important. We have not presented recent results of such investigations because the bare nucleon–nucleon forces within nuclear physics represent a discipline all its own involving intermediate and high energy physics [BJ 76a]. We have discussed some of the more conventional phenomenological ansätze (which usually involve between 40 and 50 fit parameters), but which achieve perfect reproduction of the nucleon–nucleon scattering data. However, this fit gives us information only about the on-shell behavior of the nuclear force, the off-shell part which enters into many-nucleon systems remaining indeterminate (see, for instance, [SS 76]). Three-body calculations show, however, that the currently used two-body interactions like the Hamada–Johnston potential or the Reid soft core potential in their off-shell behavior should not be very different from one another. We thus have phenomenological bare nucleon–nucleon potentials which are, in fact, quite reliable.

These potentials have, however, the inconvenient feature of being very repulsive at short distances and they can thus not be directly applied in nuclear structure calculations. Many-body theory teaches us to use effective forces instead of bare ones, the former being already an infinite sum of the latter. This procedure not only sums up higher order many-body effects in a consistent manner, but at the same time gets rid of the hard core problem, since it turns out that the effective forces are well-behaved. We shall see in the next chapter that the application of the concept of effective forces is quite successful for the calculation of ground state properties of nuclei, although the most advanced purely microscopic calculations in this field (consistently taking three-body correlations into account [KLZ 78]) are still not able to get quantitative agreement with experimental binding energies. In all other calculations, some phenomenol-

ogy enters at a certain step. Nevertheless, we can say that for the ground state properties the theory gives very good results.

Far less satisfying is the situation for the case of effective forces between valence nucleons. For example, shell model calculations within a two-particle model space and with a phenomenological two-body force are able to reproduce the low-lying spectrum of, for example, ^{18}O very well, and we may suppose that this success can be microscopically justified by calculating the effective force between the two valence nucleons. Unfortunately, these attempts have met with only partial success and show that at a certain stage in the perturbative expansion of the interaction, good agreement with experiment is achieved; but going further with the expansion worsens the results again. Until now, the reasons for this agreement at an intermediate step have not been explained.

After these numerically very involved theories of effective interactions with their moderate success, in Section 4.4 we passed to the treatment of the phenomenological effective forces. There the situation is quite satisfying. A number of phenomenological ansätze exist which have been very successful in explaining much more nuclear data than there are fit parameters. One of these forces is the Skyrme force, whose success for the ground state properties of nuclei will be discussed in more detail in the next chaper. For nuclear structure calculations we have to mention the very simple but highly successful pairing-plus-quadrupole force. In the lead region the quasi-particle concept seems to work well, as shown by the quite impressive success of calculations using the Migdal force. But other very simple forces, like the surface-delta force, also do surprisingly well. In view of their successes, it appears to be a difficult but challenging task to give an explanation from a microscopic point of view.